Dynamic Oligopoly and Innovation

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Abstract

We develop and estimate a many-firm endogenous growth model in which firms make forward-looking R&D decisions while interacting through empirically measured product-market rivalry and technology-spillover networks. The model's linear-quadratic structure reduces the dynamic oligopoly to a coupled system of Riccati equations, which we solve for the 2017 cross section of 757 publicly listed U.S. patenting firms. Gaps between social and private returns to R&D are large and heterogeneous: the median social-to-private return ratio is 1.31. A planner who controls R&D while firms remain Cournot competitors raises aggregate R&D to 231% of the competitive level, yielding a consumption-equivalent welfare gain of 0.50%. Despite wedge heterogeneity, a 33% uniform R&D subsidy delivers a 0.45% welfare gain, because the constrained planner's gain comes mainly from expanding aggregate R&D rather than reallocating it. The full planner yields a substantially larger gain of 8.97%, a margin beyond the reach of R&D subsidies.

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1 Introduction

Product-market competition and research and development (R&D) allocations are jointly determined. Firms' R&D decisions shape future productivity, product quality, and competitive positions, while current product-market rivalry feeds back into profits and the incentives to invest in R&D. The network-mediated part of these interactions operates through two pairwise channels: business stealing, which depresses competitors' profits, and technology spillovers, which raise other firms' productivity. In addition, firms do not generally appropriate the full household value generated by their innovations. Because these forces are heterogeneous across firms and firm pairs, the social-private R&D wedge is firm specific.

The two networks need not coincide: firms that compete closely in product markets may be distant in technology space, and firms that generate large spillovers may not be close product-market rivals. Yet common innovation-policy instruments, such as broad R&D subsidies and tax credits, change the price of R&D by the same proportion for every firm. The central question is which innovation margins such instruments can address, and which margins require information about firms' positions in the product-market rivalry and technology-spillover networks.

Existing endogenous growth models typically abstract from this firm-pair heterogeneity. Some models assume monopolistic competition without strategic interaction, while others allow strategic interaction among only a few firms. The difficulty is computational. In oligopoly, forward-looking R&D choices cause the dimensionality of the dynamic problem to increase rapidly with the number of strategically interacting firms. Existing quantitative frameworks therefore have had difficulty combining forward-looking strategic interaction among many firms with empirically measured product-market rivalry and technology-spillover networks. As a result, we lack firm-level measures of the social-private R&D wedges that policy should target.

This paper develops and estimates an endogenous growth model in which hundreds of firms make forward-looking R&D decisions while interacting through two empirically measured networks: a product-market rivalry network and a technology-spillover network. The contribution is twofold. Methodologically, the model's linear-quadratic structure makes a many-firm dynamic oligopoly tractable: equilibrium is characterized by a coupled system of algebraic Riccati equations rather than a high-dimensional dynamic program, so we can solve the game for the 2017 cross section of 757 U.S. publicly listed patenting firms while preserving a continuous state space and the complete firm-to-firm networks. Substantively, the estimated model delivers firm-level social-private R&D wedges and a welfare decomposition with a sharp policy message: despite large and heterogeneous wedges, a uniform R&D subsidy captures most of the gain available to a planner who controls R&D alone, while the far larger gain under the full social-planner allocation comes from reallocating product-market outcomes, a margin beyond the reach of R&D subsidies.

The model extends Pellegrino (2025)'s static product-market framework into a dynamic setting with knowledge accumulation, R&D effort, and technology spillovers. Conditional on the current knowledge-capital vector, a static product-market game under generalized hedonic linear demand maps firms' knowledge capital into quantities and profits; its key property is that gross profits are quadratic in knowledge capital. Knowledge capital accumulates linearly through own R&D effort and through the technology-spillover network. The dynamic R&D game is therefore linear-quadratic: value functions are quadratic in the state, Markov strategies are linear in it, and computation scales polynomially rather than exponentially in the number of firms. In the deterministic version, a technology transition matrix summarizes spillovers, obsolescence, and endogenous R&D and governs the evolution of the knowledge distribution.

We estimate the framework on U.S. publicly listed firms with patents over 1989–2019. Following Pellegrino (2025), we measure product-market rivalry using the product-similarity data of Hoberg and Phillips (2016); following Bloom et al. (2013) and Lucking et al. (2019), we measure technological proximity from patent-classification overlap using the DISCERN patent-firm link (Arora et al., 2024). Given these networks, the model's static equilibrium conditions recover firm-level knowledge capital from Compustat accounting data. A conditional panel relationship in recovered knowledge capital disciplines the spillover-intensity scale. The R&D efficiency and depreciation parameters are jointly calibrated to aggregate R&D intensity and instantaneous 2017 output growth. The model fits untargeted moments well: the correlation between model-implied and observed log sales is 0.96, the corresponding correlation for log R&D expenditure is 0.72, and model-implied knowledge growth evaluated at the 2010 state predicts realized 2010–2017 knowledge growth with a correlation of 0.58.

The estimated wedges are large and heterogeneous. At the observed 2017 knowledge-capital distribution, the median ratio of social to private marginal returns to R&D is 1.31, and social returns exceed private returns for 82.3% of firms with positive private and social marginal returns, consistent with aggregate underprovision. The dispersion is substantial: 22.5% of firms have ratios above 1.5, while for firms in the bottom tail the local first-order-condition measure points toward taxing rather than subsidizing marginal R&D. How much of the attainable R&D-policy gain a uniform instrument can capture is therefore a quantitative question. An exact flow-payoff decomposition separates each firm's wedge into three discounted components: an appropriability component, a rival-profit component operating through business stealing, and a closed-loop rival R&D resource-cost component. Quantitatively, the wedge is driven mainly by the net of a positive and heterogeneous appropriability component and a negative rival-profit component, while the rival R&D resource-cost component is small.

To separate the policy margins, we compare the competitive equilibrium with planner and monopoly counterfactuals. A constrained planner retains Cournot product-market conduct—the competitive mapping from knowledge capital to quantities—and chooses only the dynamic R&D effort rule. At the observed 2017 state, this raises aggregate R&D resource cost to about 231% of the competitive level, increases instantaneous output growth by 0.28 percentage points, and raises consumption-equivalent welfare by 0.50%. The corresponding constrained monopolist, which internalizes the rival-profit force but not the appropriability force, reduces aggregate R&D resource cost to roughly one half of the competitive level and lowers welfare by 1.09%. This constrained case can also be read as R&D-only common control, or an economy-wide research joint venture. Allowing product-market quantities and R&D effort to be reallocated jointly yields a much larger welfare gain under the full planner.

We then evaluate an implementable policy: a uniform R&D subsidy financed by lump-sum taxes. Household welfare peaks at a subsidy rate of 33%, which raises instantaneous output growth by 0.27 percentage points, compared with 0.28 under the constrained planner, and welfare by 0.45%, compared with 0.50%. Despite wedge heterogeneity, a uniform instrument captures roughly ninety percent of the gain attainable through R&D policy alone. The remaining welfare gap with Cournot product-market conduct fixed is small. The proximity of these gains suggests that expanding aggregate R&D expenditure accounts for most of the constrained planner's gain, while reallocating effort across firms plays a smaller role.

This paper contributes to three strands of the literature. First, it contributes to work on innovation incentives, appropriability, business stealing, technology spillovers, and innovation networks (Arrow, 1962; Spence, 1984; d'Aspremont and Jacquemin, 1988; Aghion et al., 2001, 2005; Acemoglu and Akcigit, 2012; Peters, 2020; Akcigit and Ates, 2021, 2023; De Ridder, 2024). Closest to our mechanism, Bloom et al. (2013) show empirically that product-market proximity and technological proximity have opposite effects on innovation incentives. Closest on the policy-network margin, König et al. (2019) analyze R&D alliance networks in which collaborations lower production costs while firms compete in product markets, and characterize welfare-maximizing R&D subsidies. Liu and Ma (2026) study how innovation networks shape the allocation of R&D across connected firms, and Cavenaile et al. (2026) develop an oligopolistic general-equilibrium Schumpeterian model with strategic interaction among a small number of firms. Relative to this literature, we embed product-market rivalry and technology spillovers in a many-firm dynamic oligopoly disciplined by firm-level and firm-pair-level data. This equilibrium structure lets us quantify the distribution of social-private R&D wedges, connect them to welfare, and distinguish the policy margins addressed by uniform subsidies from those tied to product-market reallocation.

Second, the paper relates to the dynamic oligopoly literature in empirical industrial organization (Ericson and Pakes, 1995; Besanko and Doraszelski, 2004; Doraszelski and Pakes, 2007; Weintraub et al., 2008). This literature develops Markov perfect industry dynamics and methods for handling large state spaces. We differ by exploiting a linear-quadratic structure that makes a full-scale many-firm continuous-state R&D game tractable with empirically measured product-market rivalry and technology-spillover networks.

Third, the paper contributes to research that incorporates oligopolistic competition into macroeconomic models (Neary, 2003; Atkeson and Burstein, 2008; Azar and Vives, 2021; Pellegrino, 2025; Ederer and Pellegrino, 2026; Bustamante and Pellegrino, 2026). Our product-market environment builds on Pellegrino (2025) and Ederer and Pellegrino (2026). Closely related contemporaneous work by Bustamante and Pellegrino (2026) develops a Network-Q model of corporate investment in oligopolistic product markets using the same GHL demand system. Their focus is neoclassical capital investment, firm valuation, markups, and merger counterfactuals; we instead add a technology-spillover network and forward-looking R&D investment to study how product-market rivalry and technology spillovers jointly shape growth, welfare, and innovation policy.

The remainder of the paper proceeds as follows. Section 2 constructs an endogenous growth model incorporating product-market rivalry and technology-spillover networks. Section 3 describes the data and estimation strategy. Section 4 reports the wedge measures, counterfactual allocations, and the uniform-subsidy evaluation. Section 5 concludes.

2 Model

We develop an endogenous growth model in which granular firms strategically interact in both product and technology spaces to examine the implications for firms' dynamic R&D effort, economic growth, and household welfare.

2.1 Environment

Our environment incorporates two networks: a product-market rivalry network and a technology-spillover network. There are n oligopolistic firms, indexed by $i \in \{1, 2, \ldots, n\}$; they choose output and dynamic R&D effort while interacting strategically through these networks. We interpret each firm as supplying one composite differentiated product, so firm and product indices coincide throughout. The set of firms, product-characteristic vectors, and technological-proximity matrix are fixed over time. Innovation changes firms' knowledge capital, labor productivity, and product quality, but not their locations in product or technology space. Time is continuous and infinite, $t \in [0, \infty)$.

2.1.1 Demand

We use the Generalized Hedonic Linear (GHL) demand system of Pellegrino (2025) to model product-market rivalry among many oligopolistic firms. In reduced form, the final-good aggregator is

$$Y_t = \boldsymbol{q}_t^T \boldsymbol{b}_t - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \tag{1}$$

where $\boldsymbol{q}_t \in \mathbb{R}^n$ is the vector of product quantities and $\boldsymbol{b}_t \in \mathbb{R}^n$ is the vector of product-level qualities. The product-rivalry matrix is

$$\boldsymbol{\Sigma} \equiv \alpha \boldsymbol{S} + (1 - \alpha)\boldsymbol{I},$$

where $\boldsymbol{S}$ is the product-similarity matrix implied by the GHL characteristic structure, $\boldsymbol{I}$ is the identity matrix, and $\alpha \in [0, 1]$ maps product similarity into substitutability. In the underlying GHL aggregator, α is the weight placed on common characteristics rather than idiosyncratic product characteristics, so higher α makes common-characteristic overlap translate more strongly into product-market substitutability. We normalize $S_{ii} = 1$, so $\sigma_{ii} = 1$ for every firm, and for $i \neq j$, $\sigma_{ij} = \alpha S_{ij}$. From Equation (1), a larger σ_{ij} implies a larger quadratic penalty when q_i and q_j are both high, so σ_{ij} measures the strength of product-market rivalry between firms i and j. The GHL characteristic-level construction underlying $\boldsymbol{S}$ and the reduced-form aggregator in Equation (1) is collected in Section A.1.

2.1.2 Technology

Each firm has a linear production technology in labor:

$$q_{i,t} = a_{i,t} l_{i,t} \tag{2}$$

where $a_{i,t}$ is the labor productivity of firm i at time t and $l_{i,t}$ is the production labor employed by firm i at time t.

Firms possess their own knowledge capital and use it to improve labor productivity and product quality. As shown in Equation (1), $b_{i,t}$ is the value (in final-good units) generated by the first unit of product i, so we interpret $b_{i,t}$ as product quality. Firms allocate knowledge capital between labor productivity $a_{i,t}$ and product quality $b_{i,t}$:

$$\zeta a_{i,t} + b_{i,t} = z_{i,t} \tag{3}$$

where $\zeta > 0$ governs the trade-off between improving labor productivity and product quality[1].

We model the law of motion for knowledge capital, which rises through technology spillovers from technologically similar firms and through the firm's own R&D effort. Spillovers operate through source firms' knowledge capital rather than directly through their contemporaneous R&D expenditures. Let $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{n\times n}$ denote recipient-normalized technology-spillover exposure, with entries $\tilde{\omega}_{ij}$. We use a row-receiver convention: row i describes the sources of spillovers received by firm i. Thus, $\tilde{\omega}_{ij} \geq 0$ is the weight placed on source firm j in firm i's spillover exposure. We set the diagonal to zero, $\tilde{\omega}_{ii} = 0$, and rows with positive off-diagonal exposure are normalized so that $\sum_{j\neq i} \tilde{\omega}_{ij} = 1$. The effective spillover matrix is $\mathbf{\Omega} = \beta\tilde{\mathbf{\Omega}}$, where $\beta > 0$ scales these relative exposure weights into spillover intensities.

Let $\boldsymbol{x}_t = (x_{1,t}, \dots, x_{n,t})^T \in \mathbb{R}^n$ denote R&D efforts, and let $\boldsymbol{W}_t = (W_{1,t}, \dots, W_{n,t})^T$ denote independent standard Wiener processes across firms. The control $x_{i,t}$ is an R&D effort that raises knowledge capital linearly, while the resource cost of generating this effort is quadratic, as specified in the resource constraint below. Knowledge capital follows a multi-dimensional geometric Brownian motion:

$$d\boldsymbol{z}_t = \left((\mathbf{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t\right) dt + \gamma \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t \tag{4}$$

where $\mu > 0$, $\delta > 0$, and $\gamma \geq 0$. The parameter μ governs the efficiency of R&D, γ the magnitude of shocks, and δ the depreciation rate of knowledge capital.

[1] Assuming optimal allocation of knowledge capital between labor-productivity and quality improvements is necessary for a balanced-growth-path equilibrium. Similarly, Grossman et al. (2017) and Jones and Liu (2024) obtain a balanced growth path by assuming two productivity-enhancing technologies (capital and human capital; capital productivity and automation, respectively) and allocating resources between them.

2.1.3 Resources and Household Preferences

We close the model with resource constraints and representative household preferences. The representative household supplies labor, owns firms, and consumes the final good. Production labor is supplied inelastically:

$$L = \sum_i l_{i,t}. \tag{5}$$

The final good is used for consumption and R&D investment. Following the endogenous growth literature (e.g. Acemoglu et al. 2016), we assume that the innovation elasticity for each firm is 0.5. Equivalently, the resource cost of firm-level R&D effort is quadratic:

$$C_t + \sum_i x_{i,t}^2 = Y_t$$

where C_t is consumption of the final good at time t. Finally, we assume that the representative household is risk-neutral and discounts the future at rate $\rho > 0$:

$$\mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right) C_t \, dt \right]$$

2.2 Competitive Equilibrium

We characterize a linear Markov perfect equilibrium. The key state variable is the vector of knowledge capital $\boldsymbol{z}$. Given $\boldsymbol{z}$, the static product-market game determines quantities, technology allocations, prices, and the real wage. Firms then choose Markov R&D policies that affect future knowledge capital through Equation (4).

Definition 1 (Competitive equilibrium)**.** A competitive equilibrium consists of Markov R&D policies $\{x_i(\boldsymbol{z})\}_{i=1}^n$, firm value functions $\{V^i(\boldsymbol{z})\}_{i=1}^n$, static allocations $\boldsymbol{q}(\boldsymbol{z})$, $\boldsymbol{a}(\boldsymbol{z})$, $\boldsymbol{b}(\boldsymbol{z})$, the real wage $w(\boldsymbol{z})/P$, and relative prices $\boldsymbol{p}(\boldsymbol{z})/P$ such that: (i) the final-good producer maximizes profits; (ii) firms play a static Cournot equilibrium given $\boldsymbol{z}$; (iii) each firm's R&D policy is a Markov best response in the dynamic problem; (iv) knowledge capital follows Equation (4); and (v) the labor market clears.

This structure lets us separate the problem into static and dynamic components. Conditional on the current knowledge-capital vector $\boldsymbol{z}_t$, the production and technology-allocation choices determine current quantities, prices, and gross profits but do not directly affect future knowledge capital. R&D affects current payoffs through its flow cost and affects future payoffs only through the law of motion for $\boldsymbol{z}_t$. Thus, we first solve the static product-market game as a mapping from knowledge capital into profits, and then use this mapping as the payoff flow in the dynamic R&D game.

2.2.1 Final-Good Producer's Problem

The final good is produced competitively. In each period, the final-good producer chooses the quantity of each product that maximizes profits, given prices.

$$\max_{\boldsymbol{q}_t} P_t Y_t - \boldsymbol{q}_t^T \boldsymbol{p}_t$$

where $\boldsymbol{p}_t$ is the vector of product prices at time t and P_t is the final-good price at time t. The final-good producer's profit maximization problem yields the following linear inverse demand function:

$$\frac{\boldsymbol{p}_t}{P_t} = \boldsymbol{b}_t - \boldsymbol{\Sigma}\boldsymbol{q}_t \tag{6}$$

2.2.2 Static Firm Problem

We assume that firms maximize real profits, so allocations are independent of the choice of numeraire (Azar and Vives, 2021). We characterize the static Cournot-Nash equilibrium in which each firm takes the real wage and rivals' quantities as given when choosing output and the allocation of knowledge capital between labor productivity and product quality. The static block maps the knowledge-capital vector into quantities:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t \tag{7}$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1},$$

and $\boldsymbol{J}$ is an $n \times n$ matrix whose entries are all one. The first term inside the inverse, $2\zeta\boldsymbol{J}/L$, captures the labor-cost channel, while $\boldsymbol{\Sigma}$ captures how quantities affect prices through product-market rivalry. The additional $\boldsymbol{I}$ is the standard Cournot own-price effect: because each firm recognizes that expanding its own quantity lowers its own price, the own-quantity term enters once more in the first-order condition. The derivation is in Section A.2. Lemma 1 shows that the inverse is well defined and that the static Cournot block is uniquely pinned down for each knowledge-capital vector.

Let $\pi_{i,t}$ denote firm i's real gross profit before subtracting R&D costs. In this equilibrium, the normalization $\sigma_{ii} = 1$ implies that static gross profit equals $q_{i,t}^2$. Hence the static payoff entering the dynamic R&D game is quadratic in knowledge capital:

$$\pi_{i,t} = q_{i,t}^2 = \boldsymbol{z}_t^T \boldsymbol{N}_i^T \boldsymbol{N}_i \boldsymbol{z}_t \equiv \boldsymbol{z}_t^T \boldsymbol{Q}_i \boldsymbol{z}_t$$

where $\boldsymbol{N}_i$ denotes the i-th row of $\boldsymbol{N}$, and $\boldsymbol{Q}_i \equiv \boldsymbol{N}_i^T \boldsymbol{N}_i$. This quadratic payoff representation is the static source of the model's linear-quadratic dynamic structure.

2.2.3 Dynamic R&D Game

Given a static production strategy profile, we consider a dynamic R&D game. In the dynamic game, given other players' strategies $\left\{x_{j,t}\right\}_{j \neq i, t \geq 0}$, firm i chooses R&D effort $\{x_{i,t}\}_{t \geq 0}$ to maximize the expected discounted present value of profits:

$$\max_{\left\{x_{i,t}\right\}_{t \geq 0}} \quad V^i\left(\boldsymbol{z}_0\right) \equiv \mathbb{E}_0\left[\int_0^{\infty} \exp\left(-\rho t\right)\left\{\pi_{i,t} - x_{i,t}^2\right\} dt\right]$$

while the state evolves according to Equation (4). We use the fact that the interest rate is pinned down by the household's Euler equation, $r_t = \rho$. In firm i's HJB equation, x_i is the choice variable and the other components of $\boldsymbol{x}$ are evaluated at rivals' Markov strategies. The dynamic problem gives the following HJB equation:

$$\begin{aligned} \rho V^i\left(\boldsymbol{z}\right) = \max_{x_i} \Bigg\{ & \boldsymbol{z}^T \boldsymbol{Q}_i \boldsymbol{z} - x_i^2 + V_{\boldsymbol{z}}^i\left(\boldsymbol{z}\right)\left[\left(\boldsymbol{\Omega} - \delta \boldsymbol{I}\right)\boldsymbol{z} + \mu \boldsymbol{x}\right] \\ & + \frac{\gamma^2}{2} \operatorname{Tr}\left[\operatorname{diag}\left(\boldsymbol{z}\right) V_{\boldsymbol{z}\boldsymbol{z}}^i\left(\boldsymbol{z}\right) \operatorname{diag}\left(\boldsymbol{z}\right)\right] \Bigg\}. \end{aligned} \tag{8}$$

The first two terms are firm i's net instantaneous profit, while the last two terms capture expected changes in continuation value from knowledge accumulation and idiosyncratic shocks.

This dynamic game is a linear-quadratic differential game, which significantly simplifies the analysis. The simplification follows from a standard guess-and-verify argument. Guess that the value can be expressed as a quadratic form of knowledge capital:

$$V^i\left(\boldsymbol{z}\right) = \boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}. \tag{9}$$

Because $\boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}$ depends only on the symmetric part of $\boldsymbol{X}^i$, we take $\boldsymbol{X}^i$ to be symmetric without loss of generality. The derivative algebra is collected in Section A.4.

In a linear-quadratic differential game, firms' strategies are expressed as linear functions of the state variables. Let X_i^i denote the i-th column of X^i. The first-order condition with respect to x_i gives the linear Markov strategy

$$x_i = \left(\mu X_i^i\right)^T z. \tag{10}$$

This linear-quadratic feedback policy is used throughout the quantitative analysis.

Given firms' linear strategies, the law of motion for knowledge capital z_t can be rewritten as a geometric Brownian motion with constant coefficients. Define $\widetilde{X}$ as the matrix obtained by stacking X_i^i:

$$\widetilde{X} \equiv \left[\begin{array}{ccc} X_1^1 & \cdots & X_n^n \end{array} \right]^T$$

Then, the law of motion is rewritten as

$$dz_t = \Phi z_t dt + \gamma \operatorname{diag}(z_t)\, dW_t$$

where the drift matrix Φ is defined by

$$\Phi \equiv \Omega - \delta I + \mu^2 \widetilde{X}.$$

We refer to the matrix Φ as the *technology transition matrix*. It consists of the exogenous technology-spillover matrix Ω, depreciation $-\delta I$, and the endogenous R&D matrix $\mu^2 \widetilde{X}$.

For any square matrix A, let $\mathcal{D}(A)$ denote the diagonal matrix formed from the diagonal entries of A. With independent multiplicative shocks, the diffusion term in Equation (8) contributes $\gamma^2 \mathcal{D}\left(X^i\right)$ to the quadratic coefficient. The HJB equation can be transformed into an algebraic Riccati equation, making the many-firm dynamic oligopoly problem computationally feasible. Substituting the quadratic value function and the linear policy Equation (10) into Equation (8), and collecting quadratic terms, we obtain stacked algebraic Riccati equations

$$0 = Q_i - \mu^2 X_i^i \left(X_i^i\right)^T + \left(\Phi - \frac{\rho}{2} I\right)^T X^i + X^i \left(\Phi - \frac{\rho}{2} I\right) + \gamma^2 \mathcal{D}\left(X^i\right) \tag{11}$$

for $i = 1, 2, \ldots, n$. Note that the terms $\mu^2 X_i^i \left(X_i^i\right)^T$ and Φ depend on the X^j of other firms $j \neq i$. Therefore, to solve for X^i, we need to simultaneously solve the algebraic Riccati equations for all firms.

We now define stability of the solution.

Definition 2. The solution of the stacked algebraic Riccati Equation (11) is *stabilizing* if it delivers finite expected discounted payoffs and satisfies the transversality condition

$$\lim_{T\to\infty} \mathbb{E}_0 \left[\exp(-\rho T) V^i\left(\boldsymbol{z}_T\right)\right] = 0$$

for all firms i and admissible initial states.

We keep the shock term in the model equations to state the stochastic LQ extension, but all quantitative exercises in the paper use the deterministic baseline, $\gamma = 0$. In that deterministic case, stability is equivalent to all eigenvalues of $\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}$ having negative real parts. With multiplicative shocks, the same definition requires the discounted second moments induced by the closed-loop process to be finite. This stability requirement corresponds to the standard growth-theory condition that the discount rate be sufficiently large for infinite-horizon utility to be finite.

Assumption 1. *There exists a stabilizing solution* $\left(\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^n\right)$ *of the stacked algebraic Riccati Equation* (11)*.*

Assumption 1 is an existence condition for a stabilizing solution of the coupled Riccati system, not a uniqueness claim. This distinction is standard in infinite-horizon nonzero-sum linear-quadratic differential games, where coupled Riccati equations can have multiple stabilizing solutions or none; see, e.g., Engwerda (2005).

Theorem 1 (Verification of the competitive feedback rule)**.** *Fix a solution* $\{\boldsymbol{X}^j\}_{j=1}^n$ *of Equation* (11)*. Suppose the induced closed-loop process is stabilizing in the sense of Definition 2. For each firm i, take rivals' feedback rules* $x_j(\boldsymbol{z}) = \mu(\boldsymbol{X}_j^j)^T \boldsymbol{z}$ *as given, where* $\boldsymbol{X}_j^j$ *denotes the jth column of* $\boldsymbol{X}^j$*. Then*

$$x_i(\boldsymbol{z}) = \mu(\boldsymbol{X}_i^i)^T \boldsymbol{z}$$

is firm i's optimal response among admissible controls for which the state equation is well defined, the expected discounted quadratic payoff is finite, and the boundary term satisfies

$$\liminf_{T\to\infty} \mathbb{E}_0 \left[e^{-\rho T} V^i(\boldsymbol{z}_T)\right] \geq 0.$$

Therefore, conditional on existence and stabilization, a solution of the stacked Riccati system induces a linear Markov perfect equilibrium in the stated admissible class.

The theorem verifies optimality among all controls in the stated admissible class, not merely within the class of linear feedback rules. Its proof is in Section A.4, and Section C.4 describes the numerical selection rule used in the reported counterfactuals.

We next use the closed-loop technology transition matrix $\boldsymbol{\Phi}$ to characterize the economy's deterministic balanced growth path.

2.2.4 Deterministic Balanced Growth Path

Set $\gamma = 0$ and consider the closed-loop technology system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. If $\boldsymbol{\Phi}$ has a simple real dominant eigenvalue $g > 0$ with a strictly positive right eigenvector $\bar{\boldsymbol{z}}$, then $\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}$ is a balanced growth path. Thus, the long-run growth rate is the dominant eigenvalue of the technology transition matrix, and the balanced-growth-path cross-section is the associated eigenvector.

Because the static mapping and R&D policies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g along this path. Output, consumption, and profits grow at rate $2g$ because they are quadratic in knowledge capital. The formal spectral condition, proof, growth-rate table, and partial-equilibrium diagram are collected in Section A.6. The counterfactuals below start from the observed 2017 knowledge-capital distribution, so this balanced-growth-path characterization is a long-run implication of the solved dynamic system rather than the basis for the main quantitative comparisons.

2.2.5 Household Welfare

Once the equilibrium is solved, household welfare can be computed for any initial knowledge-capital distribution with minimal additional cost. Let $V(\boldsymbol{z}_0)$ denote household welfare in the competitive equilibrium:

$$V(\boldsymbol{z}_0) = \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ Y_t - \sum_i x_{i,t}^2 \right\} dt \middle| \boldsymbol{z}_0 \right].$$

Total output is

$$Y_t = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$$

where

$$\boldsymbol{Q} = \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N}$$

(see Section A.3 for the derivation). The matrix $\boldsymbol{Q}$ is the output-flow matrix, not the sum of the firm payoff matrices: $\sum_i \boldsymbol{Q}_i = \boldsymbol{N}^T \boldsymbol{N}$. Thus, $\boldsymbol{Q} - \boldsymbol{N}^T \boldsymbol{N}$ is the part of static output flow not captured by aggregate gross producer profits, the static counterpart of the appropriability force in the wedge decomposition below. After firms' equilibrium policies are fixed, the remaining valuation equation contains no controls. Because the closed-loop process and the payoff are linear-quadratic, household welfare has the form $V(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X} \boldsymbol{z}$. Substituting this quadratic form into the valuation equation gives the Lyapunov equation

$$0 = \boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^T \widetilde{\boldsymbol{X}} + \boldsymbol{X} \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{X} + \gamma^2 \mathcal{D} \left(\boldsymbol{X} \right) \tag{12}$$

Therefore, $\boldsymbol{X}$ is obtained as the solution of Equation (12), and welfare in the competitive equilibrium is $V(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X} \boldsymbol{z}_0$. The valuation equation and producer-value analogue are derived in Section A.5.

2.2.6 Producer Value

We also characterize aggregate producer value in the competitive equilibrium. This object provides a measure of producers' dynamic value that can be compared with household welfare and with the counterfactual allocations below. Aggregate real profit is

$$\sum_i \pi_{i,t} = \boldsymbol{q}_t^T \boldsymbol{q}_t = \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t.$$

Relative to the household-welfare calculation in Section 2.2.5, aggregate producer value is obtained by solving a modified version of Equation (12) in which $\boldsymbol{Q}$ is replaced with $\boldsymbol{N}^T \boldsymbol{N}$. Denoting the solution by $\boldsymbol{X}^P$, aggregate producer value is $V^P(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^P \boldsymbol{z}_0$.

2.3 Planner, Monopoly, and Constrained Allocations

We use counterfactual allocations to decompose distortions along two margins: the static product-market mapping and the controller of the dynamic R&D effort rule. The scenario labels are two-letter codes: the first letter denotes the static product-market mapping, and the second denotes the R&D-effort controller. The label C denotes competition, S the social planner, and M monopoly. Table 1 summarizes the allocations defined in this section.

We first analyze the allocation chosen by a benevolent social planner who aims to maximize household welfare. We then consider the monopoly case, in which a monopolist determines production and R&D effort to maximize aggregate producer value. Finally, we define constrained counterfactuals that retain the competitive static product-market mapping but replace decentralized R&D effort choices with constrained-planner or constrained-monopolist control.

2.3.1 Social Planner

To analyze the socially optimal allocation of R&D effort and its implications for instantaneous 2017 output growth and household welfare, we formulate the social planner's problem.

Table 1: Counterfactual Allocations in the Model

Scenario	Static mapping	R&D-effort controller	Flow payoff before R&D resource cost
CC	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value	$\boldsymbol{z}^T \boldsymbol{Q}_i \boldsymbol{z}$
CS	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Constrained planner maximizes household welfare	$\boldsymbol{z}^T \boldsymbol{Q} \boldsymbol{z}$
CM	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}$
CC+s	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value with private R&D cost $(1-s)x_i^2$	$\boldsymbol{z}^T \boldsymbol{Q}_i \boldsymbol{z}$
SS	$\boldsymbol{q} = \boldsymbol{N}^*\boldsymbol{z}$	Social planner maximizes household welfare	$\frac{1}{2}\boldsymbol{z}^T \boldsymbol{N}^* \boldsymbol{z}$
MM	$\boldsymbol{q} = \boldsymbol{N}^M\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T \boldsymbol{P}^M \boldsymbol{z}$

Notes: Each dynamic objective subtracts R&D resource costs from the flow payoff shown in the last column. In CC+s, firms face the private cost $(1-s)x_i^2$; subsidy payments are lump-sum financed, so household welfare subtracts the full resource cost $\boldsymbol{x}^T\boldsymbol{x}$. The constrained scenarios CS and CM retain the competitive static mapping, $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$, and change only the controller of the R&D effort vector. Thus Cournot conduct is unchanged, although quantities evolve endogenously with knowledge capital. CM is an R&D-only common-control case in which a central controller chooses R&D effort to maximize aggregate producer value.

This full social-planner allocation corresponds to scenario SS in Table 1. The social planner aims to maximize the representative household's lifetime utility, subject to technological constraints. As in the competitive equilibrium, the social planner's problem can be separated into a static problem and a dynamic problem.

Static Problem First, we characterize the static allocation in the social planner's problem. The social planner maximizes final-good output given the vector of knowledge capital $\boldsymbol{z}_t$ in each period:

$$\max_{\boldsymbol{a}_t, \boldsymbol{b}_t, \boldsymbol{l}_t, \boldsymbol{q}_t} Y_t = \boldsymbol{q}_t^T \boldsymbol{b}_t - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t$$

subject to

$$\begin{aligned} q_{i,t} &= a_{i,t} l_{i,t} \quad i = \{1, 2, \ldots, n\} \\ z_{i,t} &= \zeta a_{i,t} + b_{i,t} \quad i = \{1, 2, \ldots, n\} \\ L &= \sum_i l_{i,t} \end{aligned} \tag{13}$$

The first constraint represents the production technology of each firm, the second governs the allocation of each firm's knowledge capital between labor productivity and product quality, and the third ensures the labor market clears. Solving the planner's static problem yields quantities as linear functions of knowledge capital:

$$\boldsymbol{q}_t^* = \boldsymbol{N}^* \boldsymbol{z}_t$$

and a quadratic representation of output in knowledge capital

$$Y_t^* = \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t$$

where

$$\boldsymbol{N}^* = \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right)^{-1}$$

(see Section B.1).

Dynamic Problem Next, we characterize the dynamic allocation in the social planner's problem. Given the static planner output coefficient, the planner chooses the R&D effort vector to maximize

$$V^{SS}(\boldsymbol{z}_0) \equiv \max_{\{x_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to the knowledge accumulation law in Equation (4). This is again a linear-quadratic control problem, so the optimal R&D effort rule is linear in knowledge capital:

$$\boldsymbol{x} = \mu \boldsymbol{X}^{SS} \boldsymbol{z} \tag{14}$$

and the induced law of motion can be written as

$$d\boldsymbol{z}_t = \boldsymbol{\Phi}^{SS} \boldsymbol{z}_t dt + \gamma \, \mathrm{diag}(\boldsymbol{z}_t) \, d\boldsymbol{W}_t$$

where

$$\boldsymbol{\Phi}^{SS} \equiv \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \boldsymbol{X}^{SS}$$

Substituting the quadratic planner value and the linear R&D rule into the planner's HJB yields a single algebraic Riccati equation:

$$0 = \frac{1}{2}\boldsymbol{N}^* - \mu^2 \left(\boldsymbol{X}^{SS}\right)^2 + \boldsymbol{X}^{SS}\left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2}\boldsymbol{I}\right) + \left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^{SS} + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^{SS}\right) \tag{15}$$

Therefore, $\boldsymbol{X}^{SS}$ is obtained as the solution of Equation (15), and household welfare under the optimal allocation is $V^{SS}(z_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{SS} \boldsymbol{z}_0$. The HJB and derivative steps are in Section B.2.

Unlike the competitive equilibrium, which requires one Riccati equation per firm, the social-planner problem requires only a single Riccati equation.

2.3.2 Monopoly

As a second counterfactual scenario, we consider the monopoly case, in which a single monopolist controls both product-market production and R&D effort. This full monopoly allocation corresponds to scenario MM. The monopolist determines all production and R&D effort to maximize aggregate producer value. In the static product-market block, the monopolist chooses $\boldsymbol{a}_t$, $\boldsymbol{b}_t$, and $\boldsymbol{q}_t$ to maximize aggregate real profit

$$\Pi_t \equiv \frac{\boldsymbol{q}_t^T \boldsymbol{p}_t}{P_t} - \frac{w_t}{P_t} \sum_i l_{i,t}.$$

Solving the monopolist's static problem yields

$$\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1},$$

and aggregate firm profit can be written as

$$\Pi_t^M = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.$$

The superscript P denotes producer value, distinguishing the monopolist's objective from household welfare under the induced policy. The monopolist's dynamic producer-value problem is therefore

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right)\left\{\boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t\right\} dt\right],$$

subject to the knowledge accumulation law. This problem is solved by replacing $\boldsymbol{N}^*/2$ with $\boldsymbol{P}^M$ in Equation (15). Let $\boldsymbol{X}^{P,MM}$ denote the solution. The monopoly R&D effort rule is then $\boldsymbol{x}_t = \mu \boldsymbol{X}^{P,MM} \boldsymbol{z}_t$. Because the monopolist maximizes producer value rather than household welfare, the household welfare reported for MM is computed separately from a Lyapunov equation using the monopoly output matrix $\boldsymbol{Q}^M$ and the technology transition induced by $\boldsymbol{X}^{P,MM}$. For the derivation of $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, $\boldsymbol{Q}^M$, and the welfare equation, see Section B.3.

2.3.3 Separating Product-Market and R&D Control

We also consider counterfactual scenarios in which control of production differs from control of R&D effort. These constrained counterfactuals retain the competitive static mapping $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$ and change only the controller of the dynamic R&D effort rule. They are useful because they separate innovation incentives from contemporaneous product-market reallocation.

First, consider scenario CS, in which firms competitively determine production quantities, and given this static competitive equilibrium, a benevolent constrained planner then selects the optimal R&D effort rule. Starting from an initial knowledge-capital distribution $\boldsymbol{z}_0$, the constrained planner maximizes the value

$$V^{CS}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^{\infty} \exp\left(-\rho t\right) \left\{ \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to

$$d\boldsymbol{z}_t = \left((\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z}_t + \mu \boldsymbol{x}_t \right) dt + \gamma \, \mathrm{diag}\left(\boldsymbol{z}_t\right) d\boldsymbol{W}_t.$$

Unlike in Section 2.3.1, the constrained planner's gross instantaneous return is $\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$ rather than $\frac{1}{2}\boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t$. Therefore, we obtain the constrained planner's allocation by solving a modified version of Equation (15), in which $\boldsymbol{N}^*/2$ is replaced with $\boldsymbol{Q}$. Denoting the solution of the modified Riccati equation by $\boldsymbol{X}^{CS}$, household welfare under the constrained-planner allocation is given by $V^{CS}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{CS} \boldsymbol{z}_0$.

The constrained monopolist in scenario CM is defined analogously, except that the dynamic controller maximizes aggregate producer value rather than household welfare. Given the competitive static mapping, $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$, aggregate real profit is $\boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t$. Hence the constrained monopolist chooses R&D effort to maximize

$$V^{P,CM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^{\infty} \exp\left(-\rho t\right) \left\{ \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to the same knowledge accumulation law. This allocation is obtained by replacing $\boldsymbol{N}^*/2$ with $\boldsymbol{N}^T\boldsymbol{N}$ in Equation (15). Let $\boldsymbol{X}^{P,CM}$ denote the resulting producer-value coefficient, so the induced R&D effort rule is $\boldsymbol{x}_t = \mu \boldsymbol{X}^{P,CM} \boldsymbol{z}_t$. Household welfare under this constrained-monopolist effort rule is then computed from the same Lyapunov equation used in Section 2.2.5, with the competitive output matrix $\boldsymbol{Q}$ and the technology transition induced by $\boldsymbol{X}^{P,CM}$. Denoting the resulting welfare coefficient by $\boldsymbol{X}^{CM}$, household welfare is $V^{CM}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{CM} \boldsymbol{z}_0$.

In the numerical analysis, we also compare these planner and monopoly allocations with the uniform-subsidy equilibrium CC+s.

3 Data and Estimation

We map the model's product-market rivalry and technology-spillover networks, knowledge capital, and dynamic parameters to observed data. We use a 1989–2019 panel to construct annual networks, recover firm-level knowledge capital, and discipline the spillover-intensity scale. The quantitative baseline and counterfactuals are solved at the observed 2017 cross section of 757 firms.

3.1 Sample Construction and Empirical Mapping

For each year, we retain firms covered by Compustat and the Hoberg–Phillips product-similarity data that have positive gross profits and at least ten DISCERN-linked patents in the five-year window used to measure technology shares. The resulting annual samples range from 418 firms in 1989 to 802 firms in 1998, and the 2017 quantitative baseline contains 757 firms. The 2017 baseline captures most activity in the relevant patenting-public-firm universe: among 2017 Compustat firms with positive gross profits and at least one DISCERN-linked patent in the 2015–2019 grant window, the baseline firms account for about 91% of sales, 98% of observed R&D, 94% of gross profits, and 99.5% of linked patents. Table 4 summarizes the estimated parameters and their sources.

Table 2 summarizes the empirical procedure: selecting the sample, constructing the two networks, recovering firm-level quantities and knowledge capital from Compustat variables, and estimating or calibrating the remaining dynamic parameters.

3.2 Product-Market Proximity and Substitutability

Following Pellegrino (2025), we measure the product-similarity matrix $\boldsymbol{S}$ using the Hoberg-Phillips 10-K text similarity data (Hoberg and Phillips, 2016). These cosine-similarity scores predict competitive relationships and provide the empirical counterpart of $\boldsymbol{\Psi}^T\boldsymbol{\Psi}$ in the GHL microfoundation in Section A.1. The numerical analysis constructs the product-rivalry matrix $\boldsymbol{\Sigma}(\alpha) = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ before solving the static and dynamic blocks, so for $i \neq j$, $\sigma_{ij} = \alpha S_{ij}$, while the model normalization imposes $S_{ii} = \sigma_{ii} = 1$.

We take α, which governs the degree of horizontal differentiation across goods, from Pellegrino (2025). That paper disciplines α using the inverse cross-price elasticity reported by Nevo (2001) and shows that the implied price elasticities are aligned with estimates for

Table 2: Mapping Data to Model Objects

Model object	Data source or empirical object	Mapping to model
$\boldsymbol{S}$, $\boldsymbol{\Sigma}(\alpha)$	Hoberg-Phillips product similarity	Hoberg cosine similarities measure $\boldsymbol{S}$; the solver uses $\boldsymbol{\Sigma} = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$, with diagonal entries normalized to one.
$\tilde{\boldsymbol{\Omega}}$	DISCERN patents and CPC groups	Jaffe overlaps are computed from five-year patent shares; the diagonal is set to zero and rows are normalized by recipient.
$\boldsymbol{q}$	Gross profits, revt − cogs	The static first-order condition implies $\pi_i = q_i^2$, so quantities are recovered as $q_i = \sqrt{\pi_i}$.
ζ/L	Aggregate COGS and recovered quantities	The aggregate production-cost condition identifies the ratio ζ/L.
$\boldsymbol{z}$	Recovered $\boldsymbol{q}$, $\boldsymbol{\Sigma}$, and ζ/L	The static equilibrium condition recovers $\boldsymbol{z}_t = (2(\zeta/L)\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I})\boldsymbol{q}_t$.
β	Panel growth of recovered knowledge capital and R&D controls	The conditional panel relationship disciplines the spillover-intensity scale.
μ and δ	R&D expenditure and instantaneous 2017 output growth	Own-R&D productivity and depreciation are jointly calibrated in the 2017 baseline; R&D effort is endogenous in the model.

Notes: Nominal accounting variables are deflated using the GDP deflator. The effective spillover matrix is $\boldsymbol{\Omega} = \beta\tilde{\boldsymbol{\Omega}}$. Observed R&D expenditures enter the panel controls and calibration moments. The recovered 2017 knowledge-capital vector is the initial state in the numerical analysis.

automobiles, ready-to-eat cereal, and computers (Berry et al., 1995; Nevo, 2001; Goeree, 2008).

Appendix Table 11 reports a local sensitivity exercise that recovers the 2017 knowledge-capital state and recalibrates the dynamic parameters at nearby values of α.

3.3 Technological Proximity

For technological proximity, we use patent data and follow Bloom et al. (2013) and Lucking et al. (2019) to construct measures of technological proximity between firms. Specifically, we use the DISCERN 2 dataset (Arora et al., 2024), which links data on U.S. publicly listed firms from the Compustat database to their patents. Following Bloom et al. (2013), we calculate technological proximity between firms using the Jaffe measure based on Cooperative Patent Classification groups. Patent shares are computed using patents granted from $t-2$ through $t+2$ around each sample year t. We use this centered grant-year window because patent grants are dated with delay relative to the underlying inventive activity: in the DISCERN-linked patents with filing years observed, the median lag between filing year and grant year is three years, and the mean lag is 3.09 years. The window therefore smooths sparse annual patent classifications and partially offsets grant-date delay, yielding a measure of firms' persistent technological locations.

Let T_i denote the vector of firm i's patent shares across technology classes. The raw Jaffe overlap (Jaffe, 1986) between firms i and j is

$$\hat{\omega}_{ij} = \frac{T_i T_j'}{\left(T_i T_i'\right)^{1/2} \left(T_j T_j'\right)^{1/2}}.$$

We measure T_i using Cooperative Patent Classification groups and construct firm-pair overlaps from the normalized technology vectors.

As in Liu and Ma (2026), we convert these raw overlaps into recipient-normalized spillover weights. We set the diagonal entries to zero and normalize each recipient firm's row:

$$\tilde{\omega}_{ij} = \frac{\hat{\omega}_{ij}}{\sum_{k \neq i} \hat{\omega}_{ik}}, \qquad j \neq i.$$

Rows with no off-diagonal overlap are left at zero. This normalization makes β capture the common intensity of spillovers received by a firm, while $\tilde{\omega}_{ij}$ determines how that exposure is allocated across source firms. Thus, for firms with positive technology overlap, the number of technology neighbors changes the composition of received spillover exposure but not the row total before scaling by β. Although the raw Jaffe overlap is symmetric, recipient normalization generally makes $\tilde{\mathbf{\Omega}}$ asymmetric. This is the empirical counterpart of the model's row-receiver convention: $\tilde{\omega}_{ij}$ is the share of firm i's spillover exposure accounted for by source firm j, and the effective spillover matrix used in the law of motion is $\mathbf{\Omega} = \beta\tilde{\mathbf{\Omega}}$.

The product-market and technology networks are empirically distinct. In the 2017 baseline, the firm-level Spearman correlation between product-market centrality and technology-source centrality is 0.22, or 0.27 using sales-weighted destination centralities.

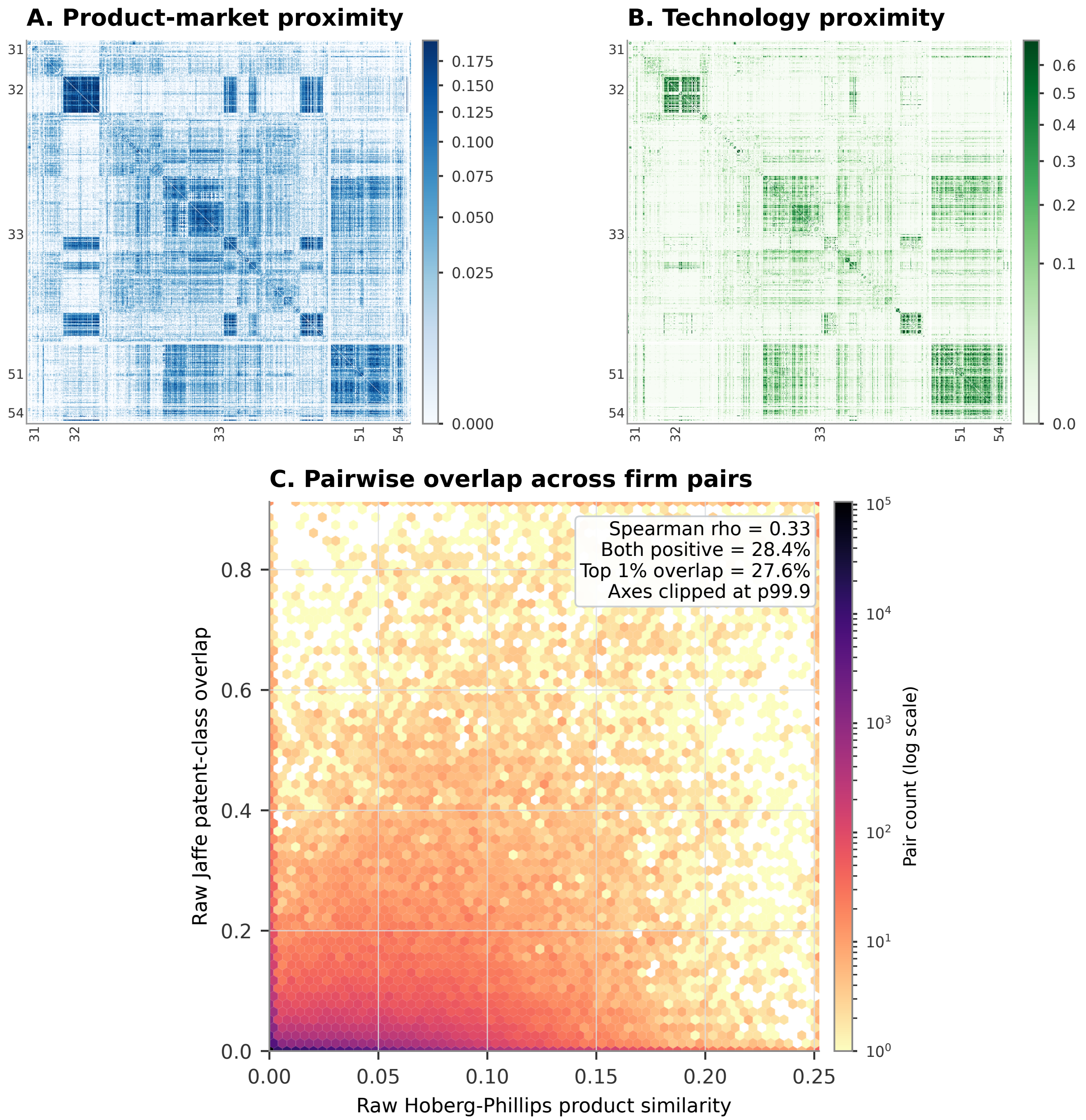
Figure 1 visualizes this distinction at the pair level. The figure uses raw product and technology proximities, before the row normalization that maps Jaffe overlap into spillover exposure.

Appendix Figure 9 shows that the product-market and technology networks are highly persistent at short horizons but decay over longer horizons. The fixed-network counterfactuals therefore describe observed-state comparisons around the 2017 network.

3.4 Knowledge Capital Distribution

Next, we recover firm-level quantities, the ratio ζ/L, and knowledge capital z_t from Compustat accounting variables. The recovery proceeds in three steps.

Figure 1: Product-Market and Technology Proximity in 2017



Notes: Panels A and B plot the same 757 firms in the 2017 baseline sample. Firms are sorted by reported detailed NAICS code and sales within detailed industry; axis labels and boundary lines mark NAICS2 sectors. Panel A reports raw Hoberg-Phillips product similarity, S_{ij}. Panel B reports raw Jaffe patent-class overlap, $\hat{\omega}_{ij}$, computed from CPC group patent shares in the centered five-year grant window. Diagonal entries are omitted from the color scale. Panel C plots the joint distribution of raw product and technology proximity across unique off-diagonal firm pairs using log-count hexagons; axes are clipped at the 99.9th percentiles for readability, while the reported statistics use all pairs. The raw technology overlap is subsequently row-normalized to construct $\tilde{\boldsymbol{\Omega}}$. The model uses $\boldsymbol{\Sigma} = \alpha \boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ for product-market rivalry and row-normalized spillover exposure $\tilde{\boldsymbol{\Omega}}$ in the knowledge law of motion.

First, we measure firm-level gross profit $\pi_{i,t}$ as total revenue (revt) minus cost of goods sold (cogs), with nominal variables deflated using the GDP deflator. In the model, the static Cournot first-order condition implies $\pi_{i,t} = q_{i,t}^2$. This identity is the reason we require positive gross profits in the sample: it lets us recover quantities as $q_{i,t} = \sqrt{\pi_{i,t}}$.

Second, we identify ζ/L from aggregate production costs. We interpret Compustat cost of goods sold as the empirical counterpart of variable production costs in the static block. Under this interpretation, the model implies

$$\frac{\zeta}{L}\boldsymbol{q}_t^T \boldsymbol{J} \boldsymbol{q}_t = \text{total production cost at time } t.$$

We calculate total production cost as the sum of cogs for all firms in the sample.

Third, we recover the knowledge-capital vector from the static equilibrium condition. Given the recovered quantities and product-rivalry matrix, Equation (7) implies

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)\boldsymbol{q}_t.$$

Because $\boldsymbol{\Sigma}$ has unit diagonal in the model, the diagonal of this recovery equation contains two identity terms: one from the unit diagonal of $\boldsymbol{\Sigma}$ and one from the Cournot own-output term in the static first-order condition. Equivalently, the empirical implementation adds $2\boldsymbol{I}$ to the stored off-diagonal product-proximity matrix. The recovered $\boldsymbol{z}_t$ is then used to estimate the law of motion and, for 2017, as the initial distribution in the numerical analysis.

3.5 Knowledge-Capital Law of Motion

In the empirical implementation, $x_{i,t}$ denotes R&D effort, measured as the square root of Compustat R&D expenditure (xrd), so that $x_{i,t}^2$ corresponds to observed R&D expenditure, consistent with the model's resource constraint. We discipline the spillover-intensity scale β using panel variation in recovered knowledge capital. Dividing the drift of the model's knowledge law of motion by $z_{i,t}$ shows that spillovers affect proportional knowledge growth through $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$, while own R&D enters through $x_{i,t}/z_{i,t}$ and depreciation enters as a common term. The depreciation component is absorbed by fixed effects and residual common drift in the panel specification, giving the annual-difference analogue:

$$\log z_{i,t+1} - \log z_{i,t} = \beta \sum_{j \neq i} \tilde{\omega}_{ij,t} \frac{z_{j,t}}{z_{i,t}} + \text{Controls}_{i,t} + \epsilon_{i,t} \tag{16}$$

Table 3: Knowledge-Capital Growth and Technology Exposure

	(1)	(2)	(3)
Technology exposure, $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$	0.039***	0.021**	0.025***
	(0.010)	(0.008)	(0.008)
Product-market exposure, $\sum_{j \neq i} s_{ij,t} z_{j,t}/z_{i,t}$		-0.008***	-0.009***
		(0.002)	(0.002)
Controls: x_i/z_i, $\log z_i$	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year/interacted FE	Year	Year	Tech-year
Observations	18,724	18,724	18,724

Notes: The dependent variable is $\Delta \log z_i$. Technology exposure uses the recipient-normalized proximity weights $\tilde{\omega}_{ij,t}$; product-market exposure uses the Hoberg–Phillips similarity scores $s_{ij,t}$. The structural spillover matrix is $\mathbf{\Omega}_t = \beta \tilde{\mathbf{\Omega}}_t$. Standard errors, two-way clustered by firm and calendar year, are in parentheses. All columns use the full sample, OLS, exact-calendar growth, firm fixed effects, x_i/z_i, and $\log z_i$. Variables are cleaned by 0.5 percent winsorization before and after exposure construction, with $\log z_i$ trimmed. Product-market exposure is included where reported. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where $z_{i,t}$ denotes firm i's knowledge capital at time t, and $\tilde{\omega}_{ij,t}$ measures the recipient-normalized technological proximity between firms i and j at time t. The controls include year and firm fixed effects, $\log z_{i,t}$, and R&D-effort intensity $x_{i,t}/z_{i,t}$. The panel specification conditions on own R&D intensity, while aggregate 2017 moments discipline μ and δ below. We also report specifications that add product-market-neighbor exposure to separate technology-neighbor exposure from product-market-neighbor exposure.

Table 3 reports estimates of Equation (16). Column (1) reports the raw technology-exposure association: the coefficient on $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$ is 0.039 and statistically significant. Column (2) adds product-market-neighbor exposure, $\sum_{j \neq i} s_{ij,t} z_{j,t}/z_{i,t}$, to separate technology exposure from product-market-neighbor exposure. The technology coefficient falls to 0.021 but remains positive and statistically significant, while the product-market exposure coefficient is negative. Column (3) replaces year fixed effects with modal-technology-section-by-year fixed effects; the technology coefficient is 0.025. Standard errors are two-way clustered by firm and calendar year in all three columns. Appendix Table 9 shows that the same three OLS specifications remain positive when the dependent variable is cumulative three-year knowledge-capital growth, $\log z_{i,t+3} - \log z_{i,t}$; the two adjusted specifications are statistically significant at the 10% level. These cumulative coefficients provide a persistence check. We use $\beta = 0.02$, the rounded adjusted one-year coefficient, as the central numerical anchor, with $\beta = 0.01$ and $\beta = 0.03$ as sensitivity cases.

3.6 Remaining Dynamic Parameters

With the spillover channel disciplined as above, we set the discount rate ρ externally and use aggregate moments to discipline the remaining dynamic parameters, the R&D coefficient μ and the depreciation rate δ. The baseline sets $\rho = 10\%$, implying relatively strong discounting of future R&D benefits. Appendix Table 12 reports sensitivity to lower discount rates.

We calibrate μ and δ jointly in the 2017 competitive-equilibrium baseline by matching two aggregate moments: R&D intensity of 2.6% and average real GDP per capita growth of 1.5% over 1989–2019. We use real GDP per capita growth because Y is real final-good output and the fixed-firm model abstracts from population growth and entry; the target therefore anchors the intensive-margin growth rate rather than nominal sales growth within the selected firms. For any candidate pair (μ, δ), we solve the 2017 competitive equilibrium at the observed knowledge-capital vector and compute the model-implied R&D intensity and the instantaneous 2017 output-growth rate, $g_Y \equiv d \log Y_t / dt$. We measure model-implied R&D intensity as aggregate R&D expenditure, $\sum_i x_i^2$, divided by labor payments plus gross operating profits. The calibration chooses μ and δ to match these two targets jointly, yielding $\mu = 0.0524$ and $\delta = 0.0112$.

Appendix Table 12 recalibrates μ and δ at $\rho \in \{5\%, 7.5\%, 10\%\}$. Lower discount rates place more weight on future R&D gains and raise their welfare value; the sensitivity exercise reports how the main quantitative comparisons change with that horizon.

Appendix Table 10 propagates the rounded beta grid through the full structural calculation. The main numerical results below use $\beta = 0.02$, with $\beta = 0.01$ and $\beta = 0.03$ as lower and upper sensitivity cases.

3.7 Model Fit and Validation

We evaluate the model's fit using both targeted and non-targeted moments. Figure 2 compares firm-level profits, sales, and R&D expenditures in the model and in the data. Profits are targeted by construction because we recover quantities from the profit identity implied by the static equilibrium. The more informative checks are therefore sales and R&D. The model reproduces the cross-sectional distribution of sales closely, with a correlation of 0.96 between model-implied and observed log sales. It also captures a substantial part of the dispersion in innovation activity: the correlation between model-implied and observed log R&D expenditures is 0.72.

Appendix Section C.1 reports descriptive cross-sectional checks relating log model-implied and observed R&D effort and the log social-to-private marginal R&D return ratio

Table 4: Parameters

Notation	Description	Value	Source
$\boldsymbol{S}$	Product similarity	–	Hoberg and Phillips (2016)
$\tilde{\boldsymbol{\Omega}}$	Technology-spillover exposure weights	–	DISCERN patents, CPC groups
α	Product similarity to rivalry	0.12	Pellegrino (2025)
β	Spillover-intensity scale	0.020	Rounded central estimate from the law of motion
γ	Std. dev. of idiosyncratic shocks	0.00	Deterministic economy in the baseline
ζ/L	Labor augmentation efficiency	0.0040	Compustat, cost of goods sold
ρ	Discount rate	0.10	Effective R&D hurdle-rate benchmark
μ	R&D coefficient	0.0524	Jointly match R&D share and growth
δ	Depreciation rate	0.0112	Jointly match R&D share and growth

Notes: Dashes indicate matrix-valued empirical objects. The baseline sets $\gamma = 0$ for the deterministic economy. Aggregate production costs identify the ratio ζ/L. The parameters μ and δ are jointly calibrated in the 2017 competitive baseline to match aggregate R&D intensity and instantaneous 2017 g_Y.

to simple product-market and technology centrality measures. The R&D columns provide reduced-form validation for the broad network forces in the model, while the return-ratio column summarizes model-implied wedge heterogeneity.

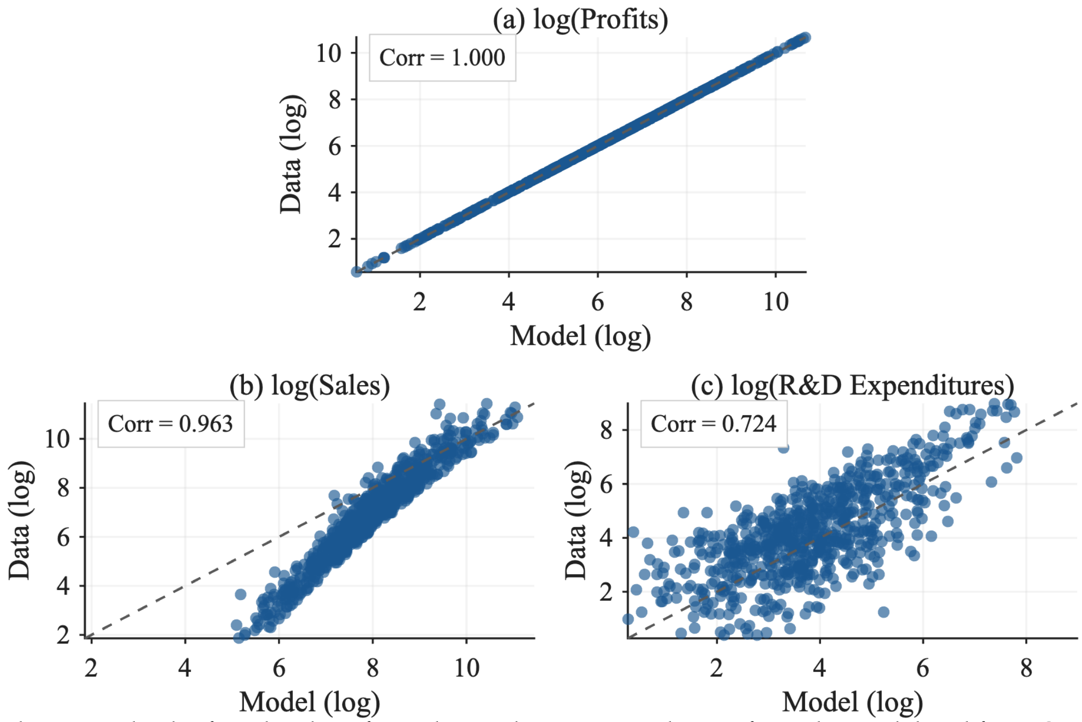
As a separate external validation, we construct a depreciated patent stock from DISCERN-linked grants. Log recovered knowledge capital is strongly correlated with the log depreciated patent stock: the overall correlation is 0.65 for count-weighted patents and 0.64 for log-citation-weighted patents, with mean annual correlations of 0.69 and 0.68, respectively.

Figure 3 compares model-implied local growth at the observed 2010 knowledge-capital distribution and networks with the subsequent annualized change in recovered knowledge capital between 2010 and 2017. The correlation is 0.579, indicating that the estimated technology-spillover network and R&D incentives generate meaningful cross-sectional variation in subsequent knowledge growth.

4 Numerical Analysis

This section reports the quantitative implications of the estimated model. Aggregate R&D resource cost in the competitive equilibrium is below the R&D-only planner level, and effort

Figure 2: Firm-Level Profits, Sales, and R&D Expenditures: Model vs. Data



Notes: These panels plot firm-level profits, sales, and R&D expenditures from the model and from Compustat. Profits in the data are measured as "Revenue – Total" (revt) minus "Cost of Goods Sold" (cogs) in Compustat. The model is calibrated so that firm-level profits match those in the data.

is misallocated across firms. A uniform subsidy captures most of the R&D-only planner gain, whereas allowing product-market reallocation produces a much larger welfare effect.

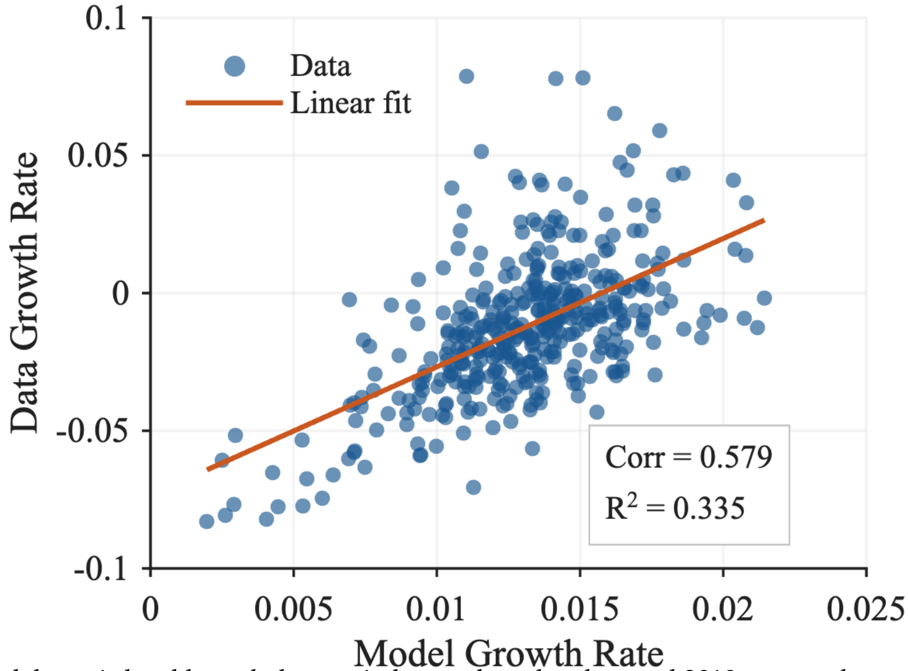
4.1 Computation and Scale

The numerical implementation follows the model's separation between static product-market mappings and dynamic R&D effort rules. For each static mapping, we construct the closed-form matrices that map knowledge capital into quantities, output, and gross producer profit. Conditional on a static block, we solve the relevant dynamic R&D effort problem and use the resulting feedback rule and value matrices for producer-value and household-welfare calculations.

The counterfactuals evaluate household welfare and the instantaneous 2017 output-growth rate at a common knowledge-capital distribution. The joint-control scenarios use closed-form static solutions. Sections C, C.5 and C.6 report the associated numerical, non-negativity, transition-path, and BGP checks.

The competitive dynamic problem solves a coupled system of Riccati equations for all firms' quadratic value functions. The implementation uses blockwise dense matrix updates,

Figure 3: Knowledge-Capital Growth Rate Comparison: Model vs. Data



Notes: The model rate is local knowledge-capital growth at the observed 2010 state and networks. The data rate is the annualized change in recovered knowledge capital between 2010 and 2017.

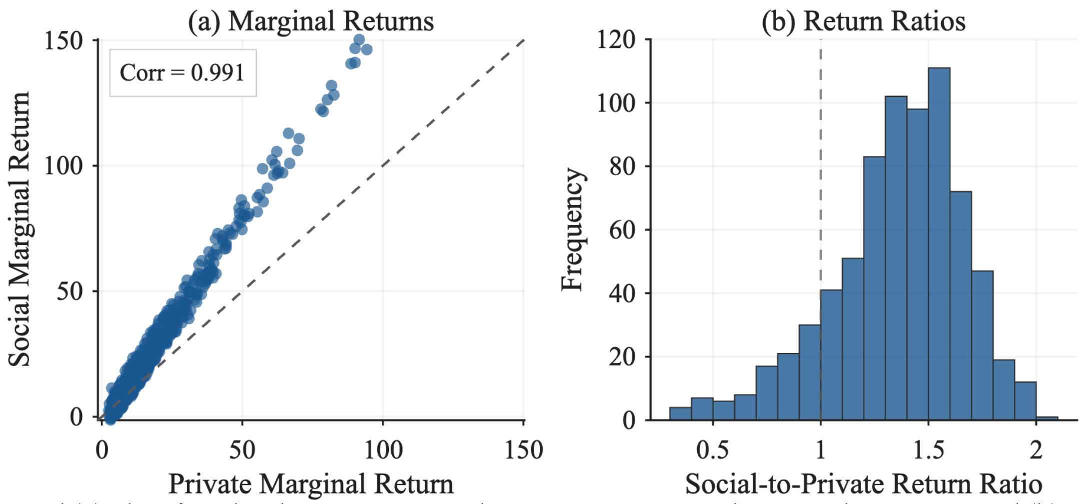
which lets the 2017 baseline solve with 757 firms while preserving a continuous state space and the full empirical product-market and technology-spillover networks. The monopoly and planner dynamic problems each reduce to a single Riccati equation because one decision maker internalizes all R&D choices. Household welfare and aggregate producer value are then computed from continuous-time Lyapunov equations. Sections C.3 and C.4 and Table 14 give the update equation, selection rule, numerical checks, and computational scale.

4.2 Social-Private R&D Wedges

We compare private and social marginal returns to R&D at the observed 2017 knowledge-capital vector. The private marginal return is the marginal increase in firm value induced by one additional unit of firm i's R&D effort before subtracting the marginal cost of that effort. Since one additional unit of x_i raises the drift of z_i by μ, this private marginal benefit is

$$\mathrm{PMR}_i(\boldsymbol{z}) \equiv \mu \frac{\partial V^i(\boldsymbol{z})}{\partial z_i} = 2\mu \left(\boldsymbol{X}_i^i\right)^\top \boldsymbol{z}.$$

Figure 4: Social-Private R&D Wedges: Marginal Returns and Return Ratios



Notes: Panel (a) plots firm-level private marginal returns against social marginal returns. Panel (b) plots the distribution of $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$ among firms with $\mathrm{PMR}_i(z) > 0$ and $\mathrm{SMR}_i(z) > 0$; the distribution trims the top and bottom 2 percent for readability, and the dashed vertical line marks equality of social and private returns. Returns are computed using the estimated networks and firm-level knowledge capital in 2017.

In vector form, $\mathrm{PMR}(z) = 2\mu\,\widetilde{\boldsymbol{X}}\,z$. Similarly, define the social marginal return as the increase in household welfare from one additional unit of firm i's R&D effort,

$$\mathrm{SMR}_i(z) \equiv \mu\,\frac{\partial V(z)}{\partial z_i} = 2\mu\,\boldsymbol{e}_i^\top \boldsymbol{X} z.$$

In vector form, $\mathrm{SMR}(z) = 2\mu\,\boldsymbol{X} z$. The social-private R&D wedge and the social-to-private return ratio are

$$\Delta_i(z) \equiv \mathrm{SMR}_i(z) - \mathrm{PMR}_i(z), \qquad \mathcal{W}_i(z) \equiv \frac{\mathrm{SMR}_i(z)}{\mathrm{PMR}_i(z)}.$$

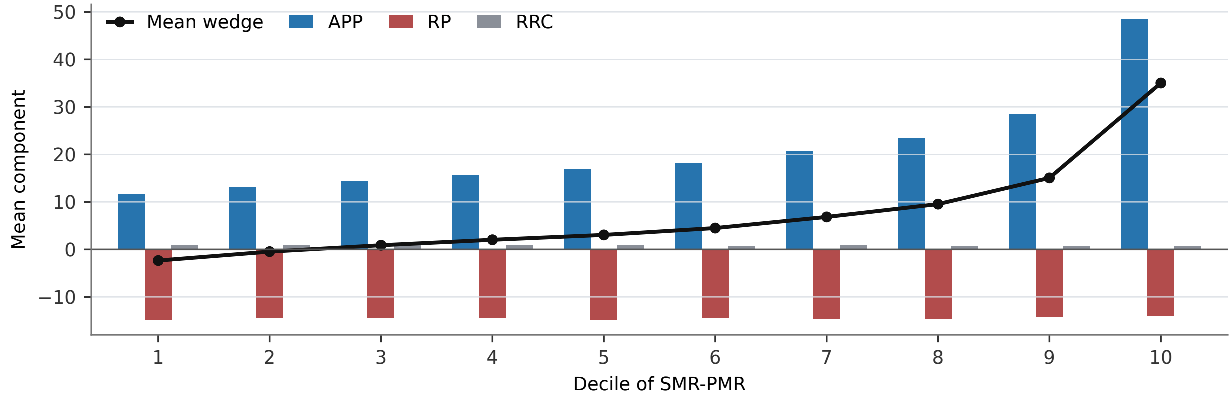
When both marginal returns are positive, we also report the local FOC-subsidy measure

$$s_i^{\mathrm{loc}}(z) \equiv 1 - \frac{\mathrm{PMR}_i(z)}{\mathrm{SMR}_i(z)}.$$

We report $s_i^{\mathrm{loc}}(z)$ as a local summary of the sign and magnitude of the observed-state wedge.

Figure 4 summarizes the firm-level wedge measures by plotting marginal-return levels and the social-to-private return-ratio distribution. The model-implied correlation between private and social marginal returns is high (0.994 after the figure's percentile trimming), but social returns tend to exceed private returns, especially among firms with high R&D returns.

Figure 5: Wedge Components by SMR-PMR Decile



Notes: Firms are sorted into deciles by the additive marginal-return wedge, $\Delta_i(z) = \mathrm{SMR}_i(z) - \mathrm{PMR}_i(z)$, in the 739-firm positive marginal-return sample at the observed 2017 state. The black line reports the decile mean of the total wedge. Bars report decile means of the exact deterministic Lyapunov components defined in Section B.5. Units are model marginal-return units. APP, RP, and RRC denote appropriability, rival-profit, and closed-loop rival R&D resource-cost components.

The wedge distribution previews the uniform-subsidy result below. The fact that social returns exceed private returns for many firms is consistent with aggregate R&D underprovision, while the dispersion indicates that the underprovision is heterogeneous across firms. Panel (b) of Figure 4 establishes the size and dispersion of the wedge. It is computed on the 739 of 757 firms with positive private and social marginal returns. In this sample, the median social-to-private return ratio is 1.31, and social returns exceed private returns for 82.3% of firms. The dispersion is economically meaningful: 22.5% of firms have ratios above 1.5 and only 0.1% exceed 2, while the bottom tail includes firms for which the local first-order-condition measure points toward taxing rather than subsidizing marginal R&D. Among firms with positive returns, the median local FOC subsidy, $1 - \mathrm{PMR}_i(z)/\mathrm{SMR}_i(z)$, is 23.4%.

Figure 5 decomposes the additive level wedge to identify the forces behind this distribution.

To interpret the figure, we decompose the wedge as

$$\Delta_i(z) = \Delta_i^{\mathrm{APP}}(z) + \Delta_i^{\mathrm{RP}}(z) + \Delta_i^{\mathrm{RRC}}(z),$$

where the three terms are the appropriability (APP), rival-profit (RP), and closed-loop rival R&D resource-cost (RRC) components of the discounted flow-payoff wedge. Technology spillovers shape all three components through the closed-loop state dynamics and the solved R&D feedback rule. The Lyapunov definitions are in Section B.5.

Economically, APP measures the part of the household value created by firm i's marginal

R&D that is not captured by aggregate producer value. It is the appropriability component of the wedge: it includes consumer surplus and other household surplus, such as labor income, generated along the induced transition path. RP measures the effect of firm i's marginal R&D on other firms' gross profits. When an innovation mainly reallocates demand or profits away from rivals, this component is negative and corresponds to business stealing. The closed-loop rival R&D resource-cost component accounts for R&D resource costs borne by other firms under their competitive feedback policies: firm i's innovation changes the future state and hence the discounted R&D resource costs borne by other firms. The signs in Figure 5 follow the wedge convention: positive entries raise social marginal returns relative to private marginal returns.

The component averages show that positive appropriability forces are partly offset by negative rival-profit effects, while RRC is small. Figure 5 sorts firms by the additive wedge, $\Delta_i(z) = \text{SMR}_i(z) - \text{PMR}_i(z)$, and reports the corresponding component means. The mean level wedge rises from −2.34 in the bottom decile to 35.03 in the top decile. This gradient is primarily an APP gradient: APP rises from 11.58 to 48.41 across the same deciles. The RP component is negative and comparatively flat, roughly between −14.80 and −14.08, so business stealing offsets a substantial part of the positive appropriability force without generating most of the cross-decile gradient. The RRC component is positive but small, between 0.70 and 0.88. Quantitatively, heterogeneity in the additive social-private wedge is mainly heterogeneity in appropriability.

4.3 Counterfactual Allocations

Table 5 compares the five allocations defined in Table 1 at the common 2017 knowledge-capital distribution. CC is the decentralized benchmark. CM and CS centralize R&D decisions while firms remain Cournot competitors; MM and SS centralize both production and R&D. The baseline instantaneous output-growth rate is 1.50%, comprising a 3.21 percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.23 point drag from obsolescence. Appendix Sections B.6 and B.7 derives and decomposes g_Y.

Table 5 reports the five allocations at the common 2017 state; output, R&D resource cost, and household welfare are normalized to CC=100.

4.3.1 Constrained R&D Control

CM and CS centralize R&D decisions while firms remain Cournot competitors. Given the same initial knowledge-capital distribution, Cournot conduct implies identical contem-

Table 5: Scenario Comparison in 2017

Scenario	Output (CC=100)	R&D resource cost (CC=100)	Instantaneous g_Y (%)	Welfare (CC=100)	Producer value share (%)
Baseline					
CC	100.00	100.00	1.50	100.00	27.08
R&D-only control, Cournot product markets					
CM	100.00	53.76	1.35	98.91	27.94
CS	100.00	231.02	1.78	100.50	25.83
Joint product-market and R&D control					
MM	96.33	62.63	1.39	95.66	43.50
SS	109.29	339.43	1.86	108.97	-5.83

Notes: All columns are evaluated at the observed 2017 knowledge-capital vector. Columns labeled CC=100 are normalized to the competitive equilibrium. R&D resource cost is $\sum_i x_i(z)^2$ under the policy used to compute growth and welfare. The instantaneous output-growth rate is $g_Y = d \log Y_t / dt$. Producer value share is $100 \times V^P/V$, net of R&D costs; its negative value in SS reflects discounted gross producer payoffs below discounted R&D costs. With linear utility, the consumption-equivalent welfare change for scenario k is $100 \times (V^k/V^{CC} - 1)$.

poraneous production quantities. Consequently, output remains $Y = 100$ across CC, CM, and CS at the observed state, although future quantities evolve with the scenario-specific knowledge-capital paths.

Both constrained controllers internalize cross-firm effects within their objective functions, but they attach different weights to the surplus created by innovation. The monopolist (CM) evaluates innovation through aggregate producer value; it internalizes losses imposed on rival producers but ignores household value not captured by aggregate producer value. Interpreted as R&D-only common control, CM describes how an economy-wide research joint venture would choose R&D while product-market conduct remains Cournot. At the observed 2017 state, it reduces aggregate R&D resource cost to 53.76, roughly one half of its competitive value. The constrained planner (CS) instead raises aggregate R&D resource cost to 231.02, more than doubling the competitive value, because it values innovation through household welfare, including surplus that firms do not appropriate. Because the product-market matrix $\boldsymbol{Q}$, spillover matrix $\boldsymbol{\Omega}$, obsolescence rate δ, and initial knowledge-capital vector $\boldsymbol{z}$ are held fixed across CC, CM, and CS, differences in instantaneous g_Y come from the endogenous R&D effort rules: g_Y falls to 1.35% under CM, 0.15 percentage points below competition, and rises to 1.78% under CS, 0.28 points above competition.

Because aggregate R&D resource cost is already below the constrained planner benchmark in the competitive equilibrium, the constrained monopolist's lower R&D effort pushes welfare down, leading to a 1.09% consumption-equivalent welfare loss relative to the competitive equilibrium outcome. The constrained planner, by contrast, lifts consumption-equivalent welfare 0.50% above competition. The producer value share moves in the opposite direction: prioritizing aggregate producer value raises the monopolist's share to 27.94%, whereas the constrained planner steers more surplus toward consumers and reduces the share to 25.83%. This constrained-planner result is the aggregate counterpart of the wedge measures above: many firms have social marginal R&D returns above private marginal returns, but the planner also reallocates effort across firms according to these heterogeneous gaps.

4.3.2 Joint Product-Market and R&D Control

In MM and SS, the monopolist or planner chooses both production and dynamic R&D effort. Comparing these allocations with CC shows the joint effect of centralizing production and innovation; CM and CS isolate the R&D-only component.

MM reduces output by 3.67% relative to competition. The monopolist chooses quantities to maximize current aggregate profit rather than household welfare, so it ignores household value not captured by aggregate producer value. By contrast, the full planner reallocates production to maximize final-good output, raising output by 9.29%.

Changes in the product market feed directly into R&D. Relative to competition, the monopolist's joint choice of production and innovation in MM reduces R&D resource cost to 62.63, roughly 37% below the competitive level. This is nevertheless higher than the 53.76 level in CM because higher markups raise the private return to new ideas. In contrast, the full planner (SS) raises R&D resource cost to 339.43, up from 231.02 in CS and about three and a half times the competitive level. This scale is consistent with the range emphasized by Bloom et al. (2013) and with the broader optimal-R&D logic in Jones and Williams (1998, 2000). The increase from CS to SS reflects the expansion of output under the planner's product-market allocation, which raises the social value of innovation.

These joint changes in production and R&D effort affect instantaneous g_Y. Under MM, g_Y is 1.39%, 0.11 percentage points below competition but 0.04 points above CM. Under SS, g_Y rises to 1.86%, 0.36 points above competition and 0.08 points above CS. The planner's reallocation accelerates knowledge accumulation, whereas the monopolist's restraint keeps growth below competition.

Letting both margins adjust magnifies the welfare stakes. Under MM, consumption-equivalent welfare falls 4.34% below competition, a much steeper loss than the 1.09% decline observed when only the R&D effort rule changes. Product-market misallocation erodes static efficiency on top of the innovation shortfall. In contrast, the full planner raises consumption-equivalent welfare by 8.97%, far exceeding the 0.50% gain from R&D-only control. Product-market reallocation delivers an immediate output gain and amplifies the social return to research.

The size of this product-market margin is close to the static oligopoly losses estimated with the same GHL demand structure. Pellegrino (2025) estimates that oligopoly lowers total surplus by about 11.5% for publicly traded corporations. Our SS allocation raises contemporaneous output by 9.29% and welfare by 8.97% in the 2017 patenting-firm sample, placing the dynamic gain in the same quantitative range while incorporating endogenous R&D and the linked Compustat-DISCERN sample.

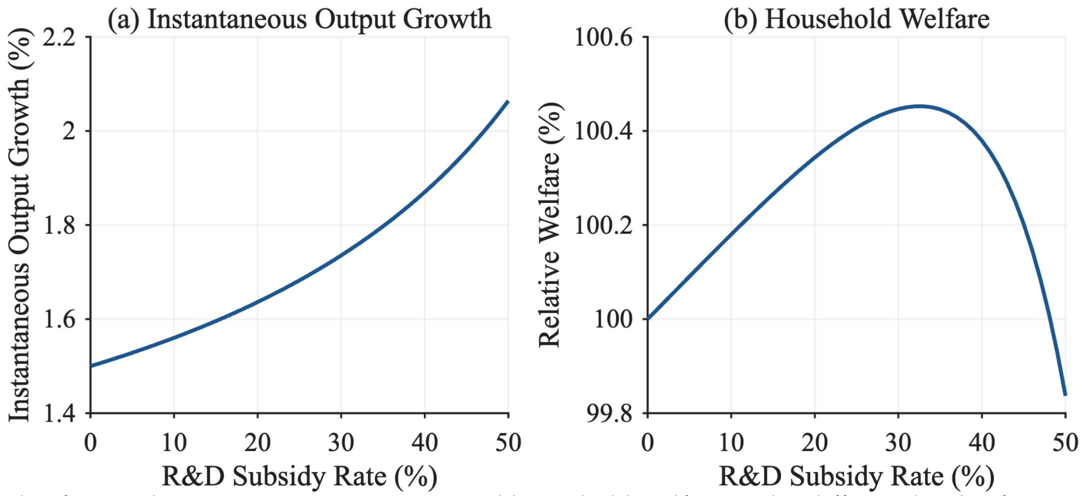
4.4 Uniform R&D Subsidy

As summarized in Table 1, a uniform R&D subsidy reduces each firm's private R&D cost from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$, where $s \in [0,1)$ is the subsidy rate. The subsidy changes firms' dynamic R&D incentives while product-market conduct remains Cournot, $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$. Given a value-function coefficient matrix, the feedback rule becomes

$$\boldsymbol{x}_t = \frac{1}{1-s}\mu\widetilde{\boldsymbol{X}}(s)\boldsymbol{z}_t.$$

We treat subsidy payments as financed by lump-sum taxes, so the subsidy changes private incentives but not the aggregate resource cost of R&D; household welfare therefore continues to subtract the full resource cost $\boldsymbol{x}_t^T\boldsymbol{x}_t$. Appendix Section B.4 records the subsidy-specific Riccati and welfare equations. In the numerical exercise, we solve the competitive equilibrium over a grid of subsidy rates, $s = 0, 0.01, \ldots, 0.50$, and choose the rate that maximizes household welfare. At the 2017 state, a uniform subsidy can improve welfare because aggregate R&D expenditure in the competitive equilibrium falls short of the constrained planner benchmark. Welfare reaches its maximum at a subsidy of 33% on this grid (Figure 6). At that rate, instantaneous g_Y rises to 1.77%, 0.27 percentage points above competition, nearly matching the constrained planner's 0.28-point increase. With Cournot product-market conduct fixed, the subsidy also captures most of the constrained planner's welfare gain: it raises consumption-equivalent welfare by 0.45%, compared with 0.50% under the constrained planner. Higher subsidies eventually lower welfare as additional R&D costs outweigh growth benefits. Appendix Figure 8 shows the associated deterministic consumption-flow paths: the subsidy and constrained-planner policies initially reduce consumption because they raise R&D resource costs, but both overtake the competitive path after about 9.5 years. The decomposition below explains why the uniform instrument comes this close to the constrained planner.

Figure 6: Instantaneous 2017 Output Growth and Household Welfare under R&D Subsidy



Notes: This figure plots instantaneous 2017 g_Y and household welfare under different levels of R&D subsidy. The product-market rivalry and technology-spillover networks, as well as the distribution of knowledge capital, are estimated using data from 2017.

4.4.1 Scale versus Allocation

How much of the constrained planner's gain comes from expanding the aggregate scale of R&D, and how much from reallocating it across firms? Both the competitive equilibrium and the constrained planner set R&D through linear feedback rules $\boldsymbol{x} = \boldsymbol{K}\boldsymbol{z}$, with $\boldsymbol{K}_C$ and $\boldsymbol{K}_S$ the competitive and constrained-planner rules. We measure the aggregate scale of a rule at the observed 2017 state by its resource cost $R(\boldsymbol{K}) = \|\boldsymbol{K}\boldsymbol{z}_0\|^2$, the same quantity that household welfare subtracts, and we hold the allocation shape fixed by scaling a rule proportionally. With $\lambda = \sqrt{R(\boldsymbol{K}_S)/R(\boldsymbol{K}_C)} = 1.52$, the rule $\lambda\boldsymbol{K}_C$ expands aggregate scale to the constrained-planner level while retaining the competitive allocation, and $\lambda^{-1}\boldsymbol{K}_S$ imposes the constrained-planner allocation at competitive scale. Table 6 evaluates the four rules spanning this grid, each scored with the same Lyapunov welfare identity used for the solved scenarios.

Scale dominates. Expanding aggregate scale alone raises consumption-equivalent welfare by 0.446%, recovering about nine-tenths of the constrained planner's 0.50% gain, whereas imposing the planner's allocation at competitive scale raises welfare by only 0.020%. The 33% uniform subsidy moves R&D resource cost to 230.91, almost exactly the CS level of 231.02, and delivers a 0.452% welfare gain. Thus, the subsidy comes close to the constrained planner because it controls the dominant aggregate-scale margin. Appendix Section B.4 reports the order-based decomposition and firm-level allocation diagnostics.

Table 6: Scale versus Allocation in the Constrained-Planner R&D Gain

Rule	Aggregate scale	Allocation shape	R&D resource cost (CC=100)	Instantaneous g_Y (%)	Welfare gain (%)
CC	CC	CC	100.00	1.50	0.000
Scale only	CS	CC	231.02	1.77	0.446
Allocation only	CC	CS	100.00	1.51	0.020
33% uniform subsidy	—	—	230.91	1.77	0.452
CS	CS	CS	231.02	1.78	0.498

Notes: All rows are evaluated at the observed 2017 knowledge-capital vector with Cournot product-market conduct, $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$, and the R&D effort rule is the linear feedback $\boldsymbol{x} = \boldsymbol{K}\boldsymbol{z}$. Aggregate scale is the R&D resource cost $R(\boldsymbol{K}) = \|\boldsymbol{K}\boldsymbol{z}_0\|^2$. Allocation shape is the normalized feedback rule: with $\lambda = \sqrt{R(\boldsymbol{K}_S)/R(\boldsymbol{K}_C)} = 1.520$, the "scale only" rule $\lambda\boldsymbol{K}_C$ holds the competitive allocation shape and moves aggregate scale to the CS level, while the "allocation only" rule $\lambda^{-1}\boldsymbol{K}_S$ holds competitive scale and adopts the planner shape. Welfare gain is $100 \times (\boldsymbol{z}_0^\top \boldsymbol{X}\boldsymbol{z}_0/\boldsymbol{z}_0^\top \boldsymbol{X}^{CC}\boldsymbol{z}_0 - 1)$. The 33% uniform subsidy is the grid-optimal implementable policy shown for reference.

4.5 Robustness

The main scenario comparison uses $\beta = 0.02$, $\alpha = 0.12$, and $\rho = 10\%$. Appendix Sections B.9 to B.11 and Tables 10 to 12 report sensitivity exercises that recalibrate the relevant dynamic parameters and resolve the corresponding counterfactuals. Across the reported values, the R&D-only planner gain remains around one half percent, the grid-optimal uniform subsidy remains close to one third, and the full-planner gain remains much larger. Varying the discount rate changes the level of discounted R&D gains, as expected, but preserves the ranking of the main scenarios.

5 Conclusion

This paper develops and estimates an endogenous growth model in which many oligopolistic firms interact through empirically measured product-market rivalry and technology-spillover networks. The linear-quadratic structure reduces the dynamic R&D game to a coupled system of Riccati equations solvable at the scale of the observed firm distribution. The same apparatus can be used in other settings in which many forward-looking agents interact through measured networks.

At the observed 2017 state, aggregate R&D resource cost in the competitive equilibrium is below the constrained planner benchmark and is misallocated across firms. The median social-to-private return ratio is 1.31 among firms with positive private and social marginal returns, with wide dispersion and a bottom tail for which the local marginal wedge points toward taxing R&D. The grid-optimal 33% uniform subsidy delivers a 0.45%

consumption-equivalent welfare gain, compared with 0.50% under the constrained planner. The proximity of these gains suggests that expanding aggregate R&D accounts for most of the constrained planner's gain, while firm-level reallocation plays a smaller role. R&D-only common control that maximizes producer value instead reduces aggregate R&D and lowers welfare. A substantially larger gain of 8.97% arises under the full planner, which jointly reallocates product-market quantities and R&D effort. This margin is beyond the reach of R&D subsidies and comparable in size to static oligopoly losses estimated with the same demand structure.

The results suggest a division of labor for innovation policy: uniform instruments for the aggregate R&D margin, and network information for diagnosing firm-level allocation differences and product-market reallocation. Extending the framework to entry and endogenous network formation is a natural next step.

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Appendix

Appendix A Model Derivations

A.1 GHL Demand Reduction

The GHL demand system begins with common characteristics and idiosyncratic product characteristics. Product i is described by an m-dimensional nonnegative vector $\boldsymbol{\psi}_i$ of common characteristics, normalized so that $\boldsymbol{\psi}_i^T \boldsymbol{\psi}_i = 1$. Stacking these vectors gives

$$\boldsymbol{\Psi} \equiv \left[\begin{array}{cccc} \boldsymbol{\psi}_1 & \boldsymbol{\psi}_2 & \cdots & \boldsymbol{\psi}_n \end{array} \right],$$

and the product-similarity matrix is

$$\boldsymbol{S} \equiv \boldsymbol{\Psi}^T \boldsymbol{\Psi}.$$

The normalization implies $S_{ii} = 1$, while for $i \neq j$, $S_{ij} = \boldsymbol{\psi}_i^T \boldsymbol{\psi}_j$ is the cosine similarity between products' common-characteristic vectors.

Let $\boldsymbol{q}_t$ be the vector of product quantities. Common and idiosyncratic characteristic quantities are

$$\boldsymbol{y}_t^C = \boldsymbol{\Psi} \boldsymbol{q}_t \tag{17}$$

and

$$\boldsymbol{y}_t^I = \boldsymbol{q}_t. \tag{18}$$

The final-good producer aggregates these characteristics according to

$$Y_t = \alpha \sum_{k=1}^{m} \left(b_{k,t}^C y_{k,t}^C - \frac{1}{2} \left(y_{k,t}^C \right)^2 \right) + (1-\alpha) \sum_{i=1}^{n} \left(b_{i,t}^I y_{i,t}^I - \frac{1}{2} \left(y_{i,t}^I \right)^2 \right), \tag{19}$$

where $\alpha \in [0, 1]$ is the weight on common characteristics. Substituting Equations (17) and (18) into Equation (19) gives

$$Y_t = \boldsymbol{q}_t^T \left[\alpha \boldsymbol{\Psi}^T \boldsymbol{b}_t^C + (1-\alpha) \boldsymbol{b}_t^I \right] - \frac{1}{2} \boldsymbol{q}_t^T \left[\alpha \boldsymbol{\Psi}^T \boldsymbol{\Psi} + (1-\alpha) \boldsymbol{I} \right] \boldsymbol{q}_t.$$

Thus the reduced-form product-quality vector and rivalry matrix are

$$\boldsymbol{b}_t \equiv \alpha \boldsymbol{\Psi}^T \boldsymbol{b}_t^C + (1-\alpha) \boldsymbol{b}_t^I, \qquad \boldsymbol{\Sigma} \equiv \alpha \boldsymbol{S} + (1-\alpha) \boldsymbol{I},$$

which yields the reduced-form aggregator in Equation (1).

A.2 Competitive Static Equilibrium

Each firm's real gross profit before subtracting R&D cost is

$$\begin{aligned}\pi_{i,t} &= \frac{p_{i,t}}{P_t} q_{i,t} - \frac{w_t}{P_t} l_{i,t} \\ &= \left(b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t},\end{aligned}$$

where the second line uses the inverse demand in Equation (6) and the production function in Equation (2). Firm i chooses $q_{i,t}$, $a_{i,t}$, and $b_{i,t}$, taking rivals' quantities, the real wage, and its own knowledge capital as given. The FOC with respect to quantity is

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2q_{i,t} - \sum_{j \neq i} \sigma_{ij} q_{j,t}. \tag{20}$$

The FOCs with respect to $a_{i,t}$ and $b_{i,t}$ imply

$$a_{i,t} = \sqrt{\frac{w_t}{\zeta P_t}} \tag{21}$$

$$b_{i,t} = z_{i,t} - \sqrt{\frac{\zeta w_t}{P_t}}. \tag{22}$$

Inserting Equation (21) into the labor-market-clearing condition in Equation (5) gives

$$\sqrt{\frac{\zeta w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \tag{23}$$

Inserting the optimality condition for technology choice Equation (21) and Equation (22) into the optimality condition for quantity choice Equation (20), and using $\sigma_{ii} = 1$ to write $2q_{i,t} + \sum_{j \neq i} \sigma_{ij} q_{j,t} = q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$, we obtain

$$z_{i,t} = 2\sqrt{\zeta \frac{w_t}{P_t}} + q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$$

Inserting the expression for real wage Equation (23), we obtain

$$z_{i,t} = q_{i,t} + \sum_j \left(2\frac{\zeta}{L} + \sigma_{ij}\right) q_{j,t}.$$

In vector form,

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right) \boldsymbol{q}_t.$$

Therefore, quantities are linear in knowledge capital:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1}.$$

Lemma 1 (Static Cournot uniqueness)**.** *Suppose $\alpha \in [0,1]$ and $\zeta/L \geq 0$. Then $2\zeta\boldsymbol{J}/L + \boldsymbol{\Sigma} + \boldsymbol{I}$ is positive definite. Hence the interior static Cournot first-order conditions have a unique solution, given by $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$.*[2]

Proof. The GHL similarity matrix is $\boldsymbol{S} = \boldsymbol{\Psi}^T\boldsymbol{\Psi}$, so it is positive semidefinite. Since $\boldsymbol{\Sigma} = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ and $\alpha \in [0,1]$, $\boldsymbol{\Sigma}$ is positive semidefinite. The all-ones matrix $\boldsymbol{J}$ is also positive semidefinite. Therefore, for any nonzero vector $\boldsymbol{v}$,

$$\boldsymbol{v}^T \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right) \boldsymbol{v} = 2\frac{\zeta}{L}\boldsymbol{v}^T\boldsymbol{J}\boldsymbol{v} + \boldsymbol{v}^T\boldsymbol{\Sigma}\boldsymbol{v} + \|\boldsymbol{v}\|^2 > 0.$$

The matrix multiplying $\boldsymbol{q}_t$ in the static first-order conditions is therefore nonsingular, so the linear system has exactly one interior solution. □

Combining the quantity FOC with $\sigma_{ii} = 1$ gives

$$b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} = q_{i,t},$$

so the equilibrium static gross profit is $\pi_{i,t} = q_{i,t}^2$.

[2] If nonnegativity of quantities were imposed in the static block, the same positive-definite matrix is a P-matrix, so the associated linear-complementarity problem also has a unique solution for each $\boldsymbol{z}_t$ (Cottle et al., 1992).

A.3 Output in Competitive Equilibrium

Let $\mathbf{1}$ denote an $n \times 1$ vector with all elements equal to 1. Rewrite Equation (22) in vector form:

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \sqrt{\zeta \frac{w_t}{P_t}} \mathbf{1}$$

Inserting Equation (23), we obtain

$$\begin{aligned} \boldsymbol{b}_t &= \boldsymbol{z}_t - \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} \\ &= \boldsymbol{z}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t . \end{aligned}$$

From Equation (7), we obtain

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t$$

Therefore, from Equation (1), the total output is given by

$$\begin{aligned} Y_t &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t \\ &= \boldsymbol{z}_t^T \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N} \boldsymbol{z}_t \\ &= \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t \end{aligned}$$

where

$$\boldsymbol{Q} \equiv \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N}$$

A.4 Competitive Dynamic R&D Derivation

For the quadratic value-function guess in Equation (9), with $\boldsymbol{X}^i$ symmetric, the first and second derivatives are

$$\frac{\partial V^i(\boldsymbol{z})}{\partial \boldsymbol{z}} = 2 \left(\boldsymbol{X}^i \boldsymbol{z} \right)^T \tag{24}$$

$$\frac{\partial^2 V^i(\boldsymbol{z})}{\partial \boldsymbol{z} \partial \boldsymbol{z}^T} = 2 \boldsymbol{X}^i . \tag{25}$$

The derivative of Equation (8) with respect to x_i is $-2x_i + \mu\left[V_z^i(z)\right]_i = 0$. Since $\left[V_z^i(z)\right]_i = 2\left(\boldsymbol{X}_i^i\right)^T \boldsymbol{z}$, the first-order condition yields the linear policy in Equation (10).

The diffusion term in Equation (8) simplifies to

$$\frac{\gamma^2}{2}\operatorname{Tr}\left[\operatorname{diag}(z)\, 2\boldsymbol{X}^i \operatorname{diag}(z)\right] = \gamma^2 \boldsymbol{z}^T \mathcal{D}\left(\boldsymbol{X}^i\right)\boldsymbol{z}.$$

Substituting Equations (9), (10), (24) and (25) into Equation (8) and collecting the quadratic terms in z gives the stacked Riccati system in Equation (11).

Proof of Theorem 1. With rivals' feedback rules fixed, firm i faces a single-agent LQ control problem. Let $V^i(z) = \boldsymbol{z}^T\boldsymbol{X}^i\boldsymbol{z}$. The Riccati equation implies that V^i satisfies firm i's HJB equation. The maximand in the HJB is strictly concave in x_i because the flow cost contains $-x_i^2$, and its first-order condition gives the feedback rule in Equation (10). For any admissible control, Itô's formula applied to $e^{-\rho t}V^i(z_t)$ on $[0, T]$ and the HJB inequality imply that the expected discounted payoff is no larger than

$$V^i(z_0) - \mathbb{E}_0\left[e^{-\rho T}V^i(z_T)\right],$$

with equality under the feedback rule above. Taking $T \to \infty$, using the admissibility condition for arbitrary controls, and using Definition 2 for the candidate feedback rule gives optimality of the candidate response. Since this argument applies to every firm, the stabilizing Riccati solution is a Markov perfect equilibrium in the stated admissible class. □

A.5 Competitive Welfare and Producer Value

After firms' equilibrium policies are fixed, household welfare is obtained from the valuation equation

$$\rho V(z) = \boldsymbol{z}^T\boldsymbol{Q}\boldsymbol{z} - \mu^2\boldsymbol{z}^T\widetilde{\boldsymbol{X}}^T\widetilde{\boldsymbol{X}}\boldsymbol{z} + V_z(z)\,\boldsymbol{\Phi}\boldsymbol{z} + \frac{\gamma^2}{2}\operatorname{Tr}\left[\operatorname{diag}(z)\, V_{zz}(z)\operatorname{diag}(z)\right].$$

The first two terms are household consumption, $Y_t - \sum_i x_{i,t}^2$, and the last two terms capture expected changes in continuation value. With $V(z) = \boldsymbol{z}^T\boldsymbol{X}\boldsymbol{z}$, substituting the derivatives from Equations (24) and (25) with $\boldsymbol{X}^i$ replaced by $\boldsymbol{X}$ gives the Lyapunov equation in Equation (12).

Table 7: Growth Rate of Variables in Deterministic Economy

Growth rate	Variables
0	$l_{i,t}$
g	$a_{i,t}, b_{i,t}, z_{i,t}, q_{i,t}, x_{i,t}, p_{i,t}/P_t$
$2g$	$\pi_{i,t}, C_t, Y_t, w_t/P_t$

Notes: Growth rates are continuous-time instantaneous rates evaluated along the balanced-growth-path equilibrium in the deterministic economy with no shocks.

Aggregate producer value in the competitive equilibrium is

$$V^P(\boldsymbol{z}_0) = \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left(\sum_i \pi_{i,t} - \sum_i x_{i,t}^2 \right) dt \middle| \boldsymbol{z}_0 \right].$$

Because aggregate real profit is $\sum_i \pi_{i,t} = \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t$, the producer-value coefficient $\boldsymbol{X}^P$ solves the same Lyapunov equation as Equation (12), replacing $\boldsymbol{Q}$ with $\boldsymbol{N}^T\boldsymbol{N}$.

A.6 Deterministic Balanced Growth Path

We characterize the existence of a balanced growth path in the deterministic economy, $\gamma = 0$, using the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. When $\gamma > 0$, the same transition matrix governs the drift of knowledge capital. Table 7 summarizes the implied growth rates of the main variables along a deterministic balanced growth path.

We impose the following spectral condition on $\boldsymbol{\Phi}$, which is sufficient for the existence of a balanced-growth-path equilibrium.

Assumption 2. *The deterministic closed-loop technology transition matrix $\boldsymbol{\Phi}$ has a simple real dominant eigenvalue $g > 0$ with an associated strictly positive right eigenvector, and all other eigenvalues have real parts strictly below g.*

This assumption imposes the spectral property needed for balanced growth: the closed-loop technology system has a unique positive long-run growth direction. It rules out cases with multiple dominant growth directions or a non-positive balanced-growth-path knowledge-capital distribution.

Definition 3. Balanced Growth Path Equilibrium: A *balanced-growth-path equilibrium* is a linear Markov perfect equilibrium such that all firms' knowledge capital grows at the same constant rate.

Proposition 1. *Consider the deterministic economy, $\gamma = 0$. Let Assumptions 1 and 2 hold, and let $\bar{\boldsymbol{z}} \gg 0$ be the right eigenvector of $\boldsymbol{\Phi}$ associated with the dominant eigenvalue g. Then, for any scalar $c > 0$, there exists a balanced-growth-path equilibrium whose knowledge-capital path is*

$$\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}.$$

Along this path, all firms' knowledge capital grows at rate g, and the cross-sectional distribution of knowledge capital is proportional to $\bar{\boldsymbol{z}}$.

Proof. Assumption 1 gives a stabilizing solution to the stacked Riccati equations and hence linear Markov R&D policies, together with the static equilibrium mappings characterized above. In the deterministic economy, these policies imply the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi}\boldsymbol{z}_t$. Since $\boldsymbol{\Phi}\bar{\boldsymbol{z}} = g\bar{\boldsymbol{z}}$, the path $\boldsymbol{z}_t = e^{gt}c\bar{\boldsymbol{z}}$ satisfies this system. Because $\bar{\boldsymbol{z}} \gg 0$ and $c > 0$, all firms' knowledge capital remains positive and grows at the common rate g. The induced static allocations and R&D policies therefore constitute a linear Markov perfect equilibrium on a balanced growth path. □

Proposition 1 implies that the equilibrium growth rate of knowledge capital is the dominant eigenvalue of $\boldsymbol{\Phi}$, and the balanced-growth-path knowledge-capital distribution is the associated eigenvector. Since $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$ and equilibrium R&D strategies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g. The static optimality conditions imply that product quality, labor productivity, and real product prices also grow at rate g, while the real wage, output, consumption, and profits grow at rate $2g$ because they are pinned down by quadratic terms in knowledge capital. In the deterministic case, Assumption 1 requires $\rho > 2g$, so discounted payoffs are finite along the balanced growth path.

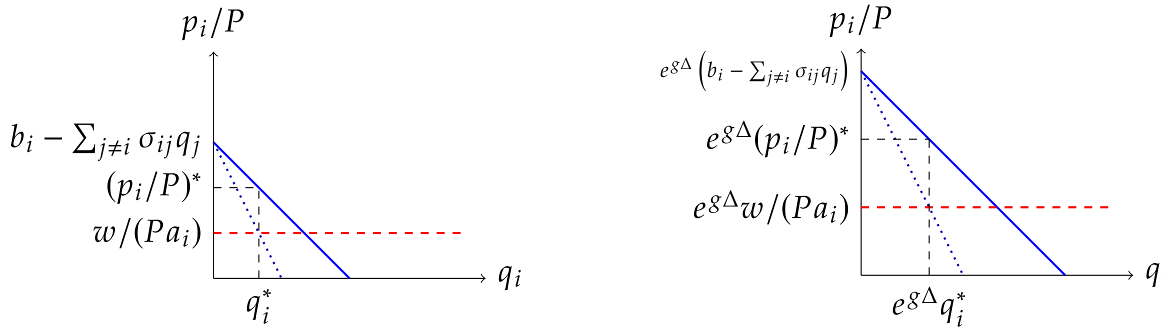
Figure 7 illustrates the mechanics of a balanced-growth-path equilibrium in the deterministic economy. In equilibrium, quality, quantity, real price, and real marginal cost grow at the same continuous-time rate g, so the partial-equilibrium diagram expands homothetically along both axes.

Appendix B Counterfactual and Wedge Derivations

B.1 Social Planner's Static Problem

We reduce the planner's static problem by substituting the knowledge-allocation constraint $b_{i,t} = z_{i,t} - \zeta a_{i,t}$ and the production technology $l_{i,t} = q_{i,t}/a_{i,t}$ into the planner's objective

Figure 7: Partial Equilibrium Diagram and Its Proportional Growth



Notes: The left panel depicts the static equilibrium for a firm's product, showing the residual demand (blue), marginal revenue (dotted line), and real marginal cost (red) curves. The right panel illustrates how the real equilibrium price, quantity, real marginal cost, and the intercept of the residual demand scale proportionally with the growth factor $\exp(g\Delta)$ over an interval of length Δ.

and labor-market constraint. Define the Lagrangian as

$$\mathcal{L} = \boldsymbol{q}_t^T \left(\boldsymbol{z}_t - \zeta \boldsymbol{a}_t\right) - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t - \lambda_t \left\{ \sum_i \frac{q_{i,t}}{a_{i,t}} - L \right\}$$

We focus on the solution in which the labor constraint binds. The FOC for $a_{i,t}$ implies $\lambda_t = \zeta a_{i,t}^2 > 0$, so the square root below is well defined. The FOC w.r.t. $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{\lambda_t}{\zeta}} \tag{26}$$

and therefore,

$$b_{i,t} = z_{i,t} - \sqrt{\zeta \lambda_t} \tag{27}$$

Inserting Equation (26) into Equation (13), we obtain

$$\sqrt{\zeta \lambda_t} = \frac{\zeta}{L} \sum_i q_{i,t}$$

Note that

$$\sqrt{\zeta \lambda_t} \mathbf{1} = \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} = \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t. \tag{28}$$

The FOC w.r.t. $\boldsymbol{q}_t$ gives

$$\boldsymbol{z}_t - \zeta \boldsymbol{a}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t = \lambda_t \begin{bmatrix} 1/a_{1,t} \\ 1/a_{2,t} \\ \vdots \\ 1/a_{n,t} \end{bmatrix}$$

Substituting Equation (26), we obtain

$$\boldsymbol{z}_t = 2\sqrt{\zeta \lambda_t} \mathbf{1} + \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28), we obtain

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t$$

Therefore,

$$\boldsymbol{q}_t = \boldsymbol{N}^* \boldsymbol{z}_t \tag{29}$$

where

$$\boldsymbol{N}^* = \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right)^{-1}$$

Because $\boldsymbol{J}$ and $\boldsymbol{\Sigma}$ are symmetric, $\boldsymbol{N}^*$ is symmetric as well. To obtain the quadratic form of output, substitute Equation (27) into Equation (1):

$$Y_t = \boldsymbol{q}_t^T \left(\boldsymbol{z}_t - \sqrt{\zeta \lambda_t} \mathbf{1} \right) - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28) together with Equation (29),

$$\begin{aligned} Y_t &= \boldsymbol{q}_t^T \left(\left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t \right) - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} \right) \boldsymbol{q}_t \\ &= \frac{1}{2} \boldsymbol{q}_t^T (\boldsymbol{N}^*)^{-1} \boldsymbol{q}_t \\ &= \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t \end{aligned}$$

B.2 Planner Dynamic R&D Derivation

The HJB equation for the social planner's dynamic problem is

$$\rho V^{SS}(\boldsymbol{z}) = \max_{\boldsymbol{x}} \left\{ \frac{1}{2}\boldsymbol{z}^T \boldsymbol{N}^* \boldsymbol{z} - \boldsymbol{x}^T \boldsymbol{x} + V_z^{SS}(\boldsymbol{z}) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z} + \mu \boldsymbol{x} \right] \right.$$
$$\left. + \frac{\gamma^2}{2} \mathrm{Tr} \left[\mathrm{diag}(\boldsymbol{z}) V_{zz}^{SS}(\boldsymbol{z}) \, \mathrm{diag}(\boldsymbol{z}) \right] \right\}. \tag{30}$$

Using the quadratic guess

$$V^{SS}(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}, \tag{31}$$

the first-order condition with respect to $\boldsymbol{x}$ is $-2\boldsymbol{x} + \mu \left(V_z^{SS}(\boldsymbol{z}) \right)^T = 0$, which gives the linear R&D rule in Equation (14). The induced transition is $\boldsymbol{\Phi}^{SS} = \boldsymbol{\Omega} + \mu^2 \boldsymbol{X}^{SS} - \delta \boldsymbol{I}$. We take the value-function coefficient matrix to be symmetric, since only its symmetric part affects $\boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}$. Thus the drift term is represented by $\boldsymbol{X}^{SS} \boldsymbol{\Phi}^{SS} + (\boldsymbol{\Phi}^{SS})^T \boldsymbol{X}^{SS}$ in quadratic form. This path-flow convention matters because the spillover matrix $\boldsymbol{\Omega}$ is generally asymmetric. Substituting the quadratic value function, the linear R&D rule, and this transition into Equation (30) gives the planner Riccati equation in Equation (15).

B.3 Monopoly Allocation and Welfare

This subsection derives the full monopoly allocation (MM) used in the counterfactual analysis. We keep the superscript M for static monopoly matrices, such as $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, and $\boldsymbol{Q}^M$, and use the superscript MM for the full scenario in which the monopolist controls both production and R&D. Since one decision maker controls the full R&D vector, the monopoly R&D policy is obtained from a single Riccati equation; household welfare under that policy is then computed separately from a Lyapunov equation.

Static Mapping

The monopolist chooses $\boldsymbol{a}_t$, $\boldsymbol{b}_t$, and $\boldsymbol{q}_t$ to maximize aggregate real profit. Differentiating Π_t with respect to $q_{i,t}$, and using the symmetry of $\boldsymbol{\Sigma}$ so that $\partial(\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t)/\partial q_{i,t} = 2 \sum_j \sigma_{ij} q_{j,t}$, gives

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2 \sum_j \sigma_{ij} q_{j,t}. \tag{32}$$

The real profit of each product is therefore

$$\begin{aligned}\pi_{i,t} &= \left(b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t}\right) q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\ &= 2\sum_j \sigma_{ij} q_{j,t} q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\ &= \sum_j \sigma_{ij} q_{i,t} q_{j,t}. \end{aligned} \tag{33}$$

The FOC with respect to $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{1}{\zeta}\frac{w_t}{P_t}}, \tag{34}$$

and therefore,

$$b_{i,t} = z_{i,t} - \sqrt{\zeta\frac{w_t}{P_t}}. \tag{35}$$

Inserting Equation (34) into the labor-market-clearing condition Equation (5) gives

$$\sqrt{\zeta\frac{w_t}{P_t}} = \frac{\zeta}{L}\sum_i q_{i,t}. \tag{36}$$

Combining Equations (32), (35) and (36) yields

$$\boldsymbol{z}_t = 2\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)\boldsymbol{q}_t, \tag{37}$$

so monopoly quantities are linear in knowledge capital:

$$\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1}. \tag{38}$$

From Equation (33), aggregate real profit is

$$\Pi_t^M = \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.$$

Output and Producer Surplus

As in the competitive equilibrium,

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \frac{\zeta}{L}\boldsymbol{J}\boldsymbol{q}_t.$$

Using Equation (37), this becomes

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t.$$

Therefore, from Equation (1), monopoly output is

$$Y_t^M = \boldsymbol{q}_t^T\left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t - \frac{1}{2}\boldsymbol{q}_t^T\boldsymbol{\Sigma}\boldsymbol{q}_t = \frac{1}{2}\boldsymbol{q}_t^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{q}_t = \boldsymbol{z}_t^T\boldsymbol{Q}^M\boldsymbol{z}_t,$$

where

$$\boldsymbol{Q}^M \equiv \frac{1}{2}(\boldsymbol{N}^M)^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{N}^M.$$

Dynamic R&D

The monopolist chooses R&D to maximize aggregate producer value:

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{\boldsymbol{z}_t^T\boldsymbol{P}^M\boldsymbol{z}_t - \boldsymbol{x}_t^T\boldsymbol{x}_t\right\}dt\right],$$

subject to the knowledge accumulation law in Equation (4). We obtain the monopoly R&D rule by solving Equation (15) with $\frac{1}{2}\boldsymbol{N}^*$ replaced by $\boldsymbol{P}^M$. Let $\boldsymbol{X}^{P,MM}$ denote the symmetric coefficient matrix in the monopolist's producer-value function. The R&D allocation is

$$\boldsymbol{x}_t = \mu\boldsymbol{X}^{P,MM}\boldsymbol{z}_t,$$

and the induced transition is

$$\boldsymbol{\Phi}^{MM} \equiv \boldsymbol{\Omega} - \delta\boldsymbol{I} + \mu^2\boldsymbol{X}^{P,MM}.$$

Household Welfare

In contrast to the social planner, the monopolist maximizes aggregate producer value rather than household utility. Hence household welfare under the monopoly policy is computed separately:

$$V^{MM}(z_0) = \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left(z_t^T \boldsymbol{Q}^M z_t - \boldsymbol{x}_t^T \boldsymbol{x}_t\right) dt \middle| z_0\right],$$

subject to $dz_t = \boldsymbol{\Phi}^{MM} z_t dt + \gamma\,\mathrm{diag}(z_t)\, d\boldsymbol{W}_t$. Guessing $V^{MM}(z) = z^T \boldsymbol{X}^{MM} z$ gives the Lyapunov equation

$$\begin{aligned} 0 = \boldsymbol{Q}^M - \mu^2 \left(\boldsymbol{X}^{P,MM}\right)^T \boldsymbol{X}^{P,MM} + \boldsymbol{X}^{MM}\left(\boldsymbol{\Phi}^{MM} - \frac{\rho}{2}\boldsymbol{I}\right) \\ + \left(\boldsymbol{\Phi}^{MM} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^{MM} + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^{MM}\right). \end{aligned} \tag{39}$$

Therefore, $\boldsymbol{X}^{MM}$ is obtained as the solution of Equation (39), and welfare in the monopoly problem is $V^{MM}(z_0) = z_0^T \boldsymbol{X}^{MM} z_0$.

B.4 Uniform R&D Subsidy Equations

This appendix records the equations used in the uniform R&D subsidy exercise. Let $s \in [0,1)$ denote a subsidy rate that reduces the private cost of R&D effort from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$. Holding the competitive static product-market mapping fixed, firm i solves

$$V^i(z_0) = \max_{\{x_{i,t}\}_{t\geq 0}} \mathbb{E}_0\left[\int_0^\infty e^{-\rho t}\left\{z_t^T \boldsymbol{Q}_i z_t - (1-s)x_{i,t}^2\right\} dt\right],$$

taking other firms' R&D rules as given. With the quadratic guess $V^i(z) = z^T \boldsymbol{X}^i z$, the first-order condition is

$$x_i = \frac{\mu}{1-s}\left(\boldsymbol{X}_i^i\right)^T z.$$

Stacking firms' policies into the subsidy-specific policy matrix $\widetilde{\boldsymbol{X}}(s)$ gives

$$\boldsymbol{x}_t = \frac{\mu}{1-s}\widetilde{\boldsymbol{X}}(s) z_t, \qquad \boldsymbol{\Phi}(s) = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \frac{\mu^2}{1-s}\widetilde{\boldsymbol{X}}(s).$$

The competitive Riccati equations under the subsidy become

$$0 = \boldsymbol{Q}_i - \frac{\mu^2}{1-s}\boldsymbol{X}_i^i\left(\boldsymbol{X}_i^i\right)^T + \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^i + \boldsymbol{X}^i\left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^i\right).$$

The subsidy changes firms' private cost but not the resource cost borne by the household.

Table 8: R&D Resource-Cost Shares by Baseline Local-Subsidy Decile

Decile	Baseline local: Median subsidy (%)	R&D resource-cost share (%): Competitive	Uniform subsidy	Planner (CS)
1	-71.8	0.5	0.4	0.1
2	-6.1	1.2	1.2	0.5
3	9.1	1.7	1.6	0.9
4	16.3	2.4	2.3	1.4
5	21.2	3.4	3.3	2.4
6	25.7	4.4	4.4	3.5
7	29.4	7.5	7.4	6.6
8	32.5	18.3	18.3	17.9
9	35.3	41.7	42.0	44.6
10	39.8	18.9	19.0	22.2

Notes: Deciles sort firms by the baseline local subsidy, $s_i^{\text{loc}}(z) = 1 - \text{PMR}_i(z)/\text{SMR}_i(z)$, among the 739 firms with positive private and social marginal returns. Negative values indicate a local tax. R&D resource-cost shares are the squared-effort shares $x_i^2/\sum_j x_j^2$ at the observed 2017 state. "Uniform" is the solved 33% competitive-subsidy equilibrium. "Planner" is the constrained-planner allocation (CS), evaluated under Cournot product-market conduct.

Hence household welfare subtracts the full cost

$$\boldsymbol{x}_t^T \boldsymbol{x}_t = \frac{\mu^2}{(1-s)^2} \boldsymbol{z}_t^T \widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) \boldsymbol{z}_t.$$

Given the equilibrium policy, household welfare is obtained from the Lyapunov equation

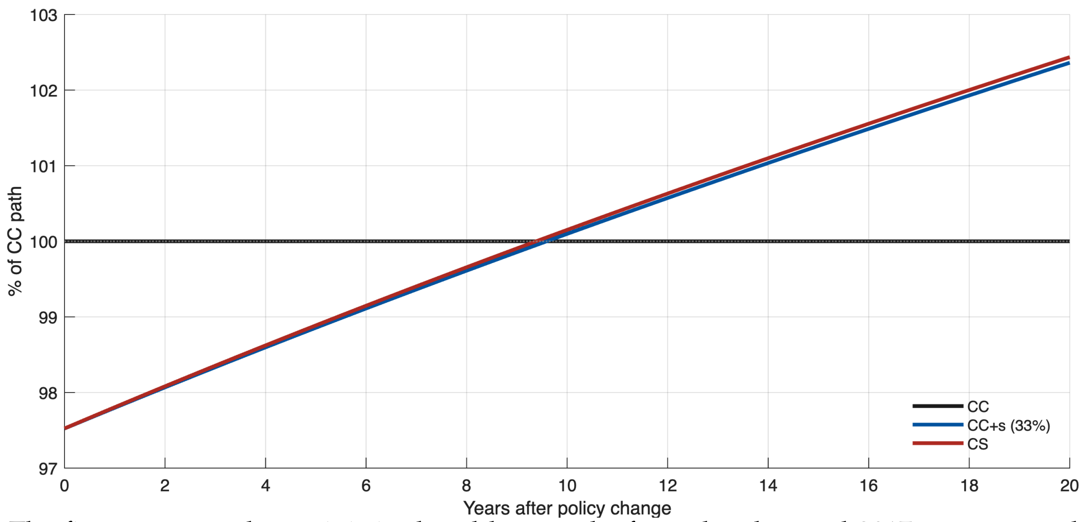
$$\begin{aligned} 0 = \boldsymbol{Q} - \frac{\mu^2}{(1-s)^2} \widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) + \boldsymbol{X}(s) \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2} \boldsymbol{I} \right) \\ + \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{X}(s) + \gamma^2 \mathcal{D} \left(\boldsymbol{X}(s) \right). \end{aligned}$$

The numerical exercise solves this system for $s = 0, 0.01, \ldots, 0.50$ and selects the value of s that maximizes $\boldsymbol{z}_0^T \boldsymbol{X}(s) \boldsymbol{z}_0$ at the 2017 knowledge-capital vector.

Table 8 reports the decile-level allocation diagnostic behind the uniform-subsidy comparison in the main text. Since the subsidy exercise changes a common proportional R&D price, the table sorts firms by the baseline local first-order-condition subsidy, $s_i^{\text{loc}}(z) = 1 - \text{PMR}_i(z)/\text{SMR}_i(z)$, rather than by the additive level wedge used in Figure 5.

Because scale and allocation interact, the fraction of the CS welfare gain assigned to scale depends on the order of the two changes. Moving scale first assigns 90% of the gain

Figure 8: Short-Run Consumption-Flow Paths under R&D Subsidy and Constrained Planner



Notes: The figure reports deterministic closed-loop paths from the observed 2017 state, normalized by the contemporaneous competitive path so that CC equals 100 at each horizon. The subsidy path uses the grid-optimal 33% uniform R&D subsidy. Subsidy payments are lump-sum financed; the initial decline reflects higher R&D resource costs.

to scale, while moving allocation first assigns 96%; the Shapley average assigns 93% to scale and 7% to allocation. At the firm level, let $r_i^k = (x_i^k)^2 / \sum_j (x_j^k)^2$ denote firm i's R&D resource-cost share under allocation k. The total-variation distance from the CS shares is 0.0629 under competition and 0.0582 under the 33% subsidy. The subsidy-induced and planner-induced changes in these shares have a correlation of 0.840. These diagnostics show that the subsidy primarily expands aggregate R&D and only modestly shifts its firm-level allocation.

B.5 Social-Private R&D Wedge Decomposition

This appendix gives the component definitions used in Figure 5. Let $\boldsymbol{e}_i$ denote the ith standard basis vector. Let $\boldsymbol{P}$ denote the aggregate competitive gross-profit matrix,

$$\boldsymbol{P} \equiv \sum_{j=1}^{n} \boldsymbol{Q}_j = \boldsymbol{N}^T \boldsymbol{N}.$$

Thus aggregate competitive gross profit is $\sum_j \pi_j(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{P} \boldsymbol{z}$. The social-private R&D wedge can be written as

$$\Delta_i(\boldsymbol{z}) = 2\mu\, \boldsymbol{e}_i^T \left(\boldsymbol{X} - \boldsymbol{X}^i \right) \boldsymbol{z}. \tag{40}$$

Define $\boldsymbol{M}_i \equiv \boldsymbol{X} - \boldsymbol{X}^i$ and

$$\boldsymbol{C}_{-i} \equiv \widetilde{\boldsymbol{X}}^T \left(\boldsymbol{I} - \boldsymbol{e}_i\boldsymbol{e}_i^T\right) \widetilde{\boldsymbol{X}}.$$

Since the competitive feedback rule is $\boldsymbol{x}^C(\boldsymbol{z}) = \mu\widetilde{\boldsymbol{X}}\boldsymbol{z}$, the quadratic form $\mu^2\boldsymbol{C}_{-i}$ gives the R&D resource costs of all firms other than i under the competitive policy. Subtracting firm i's Riccati equation from the household welfare Lyapunov equation gives

$$\boldsymbol{0} = \boldsymbol{F}_i + \boldsymbol{M}_i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{M}_i + \gamma^2\mathcal{D}(\boldsymbol{M}_i), \tag{41}$$

where the flow wedge is

$$\boldsymbol{F}_i \equiv \boldsymbol{Q} - \boldsymbol{Q}_i - \mu^2\boldsymbol{C}_{-i} = \underbrace{\boldsymbol{Q} - \boldsymbol{P}}_{\text{appropriability}} + \underbrace{\boldsymbol{P} - \boldsymbol{Q}_i}_{\text{rival profit}} - \underbrace{\mu^2\boldsymbol{C}_{-i}}_{\text{rival R\&D cost}} . \tag{42}$$

This decomposition separates the flow-payoff consequences of a marginal increase in firm i's knowledge drift under the competitive closed-loop policy. The term $\boldsymbol{Q} - \boldsymbol{P}$ is the household output value that is not captured by aggregate producer gross profit, so it is the appropriability component; as noted in Section 2.2.5, this is the static output-profit gap $\boldsymbol{Q} - \boldsymbol{N}^T\boldsymbol{N}$. The term $\boldsymbol{P} - \boldsymbol{Q}_i$ is the gross-profit flow accruing to firms other than i; the reported RP component is its marginal value with respect to z_i, which is negative when higher z_i lowers rivals' quantities. Finally, $-\mu^2\boldsymbol{C}_{-i}$ subtracts the R&D resource costs borne by other firms under their competitive feedback rules. The sign of the resulting RRC marginal component depends on the closed-loop feedback response; the positive average entries in Figure 5 indicate that the marginal state change lowers discounted rival R&D resource costs in the baseline decile groups. These components decompose flow payoffs rather than primitive network channels: technology spillovers enter through the closed-loop state dynamics and the solved R&D feedback rule, so they shape the discounted APP, RP, and RRC components rather than appearing as a separate component.

For each component $m \in \{\text{APP}, \text{RP}, \text{RRC}\}$, solve Equation (41) with the corresponding flow term

$$\boldsymbol{F}^{\text{APP}} = \boldsymbol{Q} - \boldsymbol{P}, \qquad \boldsymbol{F}_i^{\text{RP}} = \boldsymbol{P} - \boldsymbol{Q}_i, \qquad \boldsymbol{F}_i^{\text{RRC}} = -\mu^2\boldsymbol{C}_{-i}.$$

Denoting the resulting solutions by $\boldsymbol{M}^{\text{APP}}$, $\boldsymbol{M}_i^{\text{RP}}$, and $\boldsymbol{M}_i^{\text{RRC}}$, the component wedges are

$$\begin{aligned} \Delta_i^{\text{APP}}(\boldsymbol{z}) &= 2\mu\, \boldsymbol{e}_i^T \boldsymbol{M}^{\text{APP}} \boldsymbol{z}, \\ \Delta_i^{\text{RP}}(\boldsymbol{z}) &= 2\mu\, \boldsymbol{e}_i^T \boldsymbol{M}_i^{\text{RP}} \boldsymbol{z}, \\ \Delta_i^{\text{RRC}}(\boldsymbol{z}) &= 2\mu\, \boldsymbol{e}_i^T \boldsymbol{M}_i^{\text{RRC}} \boldsymbol{z}, \end{aligned} \tag{43}$$

Because Equation (41) is linear in the pair $(\boldsymbol{M}, \boldsymbol{F})$, the component solutions sum to $\boldsymbol{M}_i$, so the component wedges add up exactly to the total wedge:

$$\Delta_i(\boldsymbol{z}) = \Delta_i^{\mathrm{APP}}(\boldsymbol{z}) + \Delta_i^{\mathrm{RP}}(\boldsymbol{z}) + \Delta_i^{\mathrm{RRC}}(\boldsymbol{z}). \tag{44}$$

B.6 Instantaneous Expected Log Output Growth

This subsection derives the instantaneous expected log-output growth formula used in the baseline decomposition below. Apply Itô's lemma to the closed-loop knowledge law

$$d\boldsymbol{z}_t = \boldsymbol{\Phi}\boldsymbol{z}_t dt + \gamma \,\mathrm{diag}\,(\boldsymbol{z}_t)\, d\boldsymbol{W}_t$$

and to output $Y_t = f(\boldsymbol{z}_t) = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$. The matrix $\boldsymbol{Q}$ is symmetric by construction, and the log-growth formula below is evaluated on states with $Y_t > 0$. The required derivatives are

$$f_z\,(\boldsymbol{z}_t) = 2\boldsymbol{Q}\boldsymbol{z}_t,$$
$$f_{zz}\,(\boldsymbol{z}_t) = 2\boldsymbol{Q}.$$

Therefore,

$$\begin{aligned} dY_t &= \left\{2\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{\Phi}\boldsymbol{z}_t + \frac{1}{2}\mathrm{Tr}\left[2\gamma^2\,\mathrm{diag}\,(\boldsymbol{z}_t)\,\boldsymbol{Q}\,\mathrm{diag}\,(\boldsymbol{z}_t)\right]\right\} dt + 2\gamma \boldsymbol{z}_t^T \boldsymbol{Q}\,\mathrm{diag}\,(\boldsymbol{z}_t)\, d\boldsymbol{W}_t \\ &= \left\{\boldsymbol{z}_t^T \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T \boldsymbol{Q}\right) \boldsymbol{z}_t + \gamma^2 \sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}\right\} dt + 2\gamma \boldsymbol{z}_t^T \boldsymbol{Q}\,\mathrm{diag}\,(\boldsymbol{z}_t)\, d\boldsymbol{W}_t \end{aligned}$$

Next, applying Itô's lemma to $g\,(Y_t) = \log Y_t$ and using the diffusion term in dY_t gives

$$\begin{aligned} d\log Y_t =& \left[\frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{z}_t} + \gamma^2 \left\{\frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{z}_t} - 2\frac{\boldsymbol{z}_t^T \boldsymbol{Q}\,\mathrm{diag}\,(\boldsymbol{z}_t^2)\,\boldsymbol{Q}\boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{z}_t\right)^2}\right\}\right] dt \\ &+ \frac{2\gamma}{\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{z}_t} \boldsymbol{z}_t^T \boldsymbol{Q}\,\mathrm{diag}\,(\boldsymbol{z}_t)\, d\boldsymbol{W}_t. \end{aligned}$$

Because the technology transition matrix can be written as

$$\boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}},$$

the first drift term in log output can be decomposed as

$$\boldsymbol{Q\Phi} + \boldsymbol{\Phi}^T \boldsymbol{Q} = \left(\boldsymbol{Q\Omega} + \boldsymbol{\Omega}^T \boldsymbol{Q}\right) + \mu^2 \left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T \boldsymbol{Q}\right) - 2\delta \boldsymbol{Q}.$$

Thus, the contributions of technology spillovers, R&D, obsolescence, and the Itô term to instantaneous 2017 g_Y are

$$\frac{d \log Y_t|_{\text{spillover}}}{dt} = \frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q\Omega} + \boldsymbol{\Omega}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} \tag{45}$$

$$\frac{d \log Y_t|_{\text{R\&D}}}{dt} = \mu^2 \frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} \tag{46}$$

$$\frac{d \log Y_t|_{\text{Obsolescence}}}{dt} = -2\delta \tag{47}$$

$$\frac{d \log Y_t|_{\text{Itô}}}{dt} = \gamma^2 \left\{ \frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} - 2 \frac{\boldsymbol{z}_t^T \boldsymbol{Q} \operatorname{diag}\left(\boldsymbol{z}_t^2\right) \boldsymbol{Q} \boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t\right)^2} \right\}.$$

B.7 Baseline Instantaneous Growth Decomposition

To interpret the g_Y column in the counterfactual table, we decompose the baseline instantaneous 2017 output-growth rate into technology spillovers, obsolescence, R&D, and the contribution of shocks. The components are computed from the primitive matrices $\boldsymbol{Q}$, $\boldsymbol{\Omega}$, $\widetilde{\boldsymbol{X}}$, and the observed 2017 knowledge-capital vector so that they add up to g_Y under the baseline policy.

The baseline instantaneous 2017 output-growth rate is 1.50%, equal to the growth target used in the calibration up to numerical tolerance. It is the net of a 3.21 percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.23 point drag from knowledge obsolescence. In the baseline deterministic specification, the Itô term is zero because $\gamma = 0$. This decomposition guides the counterfactuals in the main text: changing R&D incentives affects g_Y through the endogenous R&D component, while changing product-market allocations changes the output matrix $\boldsymbol{Q}$ and hence how a given knowledge-capital distribution maps into aggregate output.

B.8 Spillover-Intensity Horizon Robustness

Table 9 reports the same three OLS spillover-intensity specifications using cumulative three-year knowledge-capital growth as the dependent variable. The estimates remain positive across the baseline controls, the product-market horse race, and the technology-section-by-

Table 9: Horizon Robustness of Spillover-Intensity Estimates

	Technology-exposure coefficient	
	$h = 1$ (annual)	$h = 3$ (cumulative)
(1) OLS + controls	0.039***	0.070***
	(0.010)	(0.023)
(2) + Product-market exposure	0.021**	0.037*
	(0.008)	(0.021)
(3) + Tech-section × year FE	0.025***	0.037*
	(0.008)	(0.020)
Observations	18,724	14,847

Notes: The dependent variable is cumulative knowledge-capital growth, $\log z_{i,t+h} - \log z_{i,t}$. Rows reproduce columns (1)–(3) of Table 3. The $h = 3$ estimates report cumulative three-year growth. Standard errors, two-way clustered by firm and calendar year, are in parentheses. Variables use the same 0.5 percent winsorization rule as in the main table. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Spillover-Intensity Sensitivity

β	μ	δ	CC g_Y components, pp			Welfare		Uniform subsidy	
			Spill.	R&D	Dep.	CS welfare	SS welfare	Subsidy rate (%)	Subsidy welfare
0.010	0.0473	0.0029	1.61	0.47	-0.58	100.37	109.99	29	100.32
0.020	0.0524	0.0112	3.21	0.52	-2.23	100.50	108.97	33	100.45
0.030	0.0572	0.0194	4.82	0.56	-3.88	100.76	108.38	37	100.68

Notes: For each reported β, μ and δ are recalibrated in the 2017 competitive equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 output growth g_Y of 1.5%. Spill., R&D, and Dep. decompose competitive-equilibrium g_Y in percentage points. CS, SS, and uniform-subsidy welfare are normalized to the competitive equilibrium at the same β. The uniform-subsidy rate is the grid-optimal R&D subsidy on the 0–50% grid.

year fixed-effect specification; the two adjusted estimates are statistically significant at the 10% level. These cumulative coefficients provide a persistence check for the one-year panel relationship.

B.9 Spillover-Intensity Sensitivity

Because β affects the counterfactuals through spillover growth, the recalibration of μ and δ, and the solved R&D policies, Table 10 propagates the rounded positive grid $\beta \in \{0.01, 0.02, 0.03\}$ through the full structural calculation.

Table 11: Product-Market Parameter Sensitivity

		Scenario outcomes (CC=100)			Uniform subsidy	
α	Median SMR/PMR	CS welfare	SS output	SS welfare	Rate (%)	Welfare (CC=100)
0.08	1.319	100.53	109.86	109.50	33	100.49
0.12	1.305	100.50	109.29	108.97	33	100.45
0.16	1.296	100.47	108.79	108.51	32	100.42

Notes: For each α, the 2017 knowledge-capital state is recovered from accounting profits using the corresponding product-market matrix, and μ and δ are recalibrated in the competitive equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 g_Y of 1.5%. The spillover parameter is fixed at $\beta = 0.02$ and the discount rate at $\rho = 0.10$. Median SMR/PMR is the median of $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$ among firms with positive private and social marginal R&D returns. CS welfare, SS output, SS welfare, and subsidy welfare are normalized to no-subsidy competition at the same α. The subsidy rate is the welfare-maximizing uniform R&D subsidy on the 0–50% grid with 1 percentage point spacing.

B.10 Product-Market Parameter Sensitivity

Table 11 varies the product-market parameter around the baseline value while keeping the central spillover parameter fixed at $\beta = 0.02$. For each value, the 2017 knowledge-capital state is recovered with the corresponding product-market matrix and the dynamic parameters are recalibrated to the same R&D-intensity and instantaneous 2017 output-growth targets. The main quantitative ordering is stable: the R&D-only planner gain remains around one half percent, the grid-optimal uniform subsidy remains close to one third, and the full-planner gain remains much larger.

B.11 Discount-Rate Sensitivity

Table 12 varies the discount rate and recalibrates the dynamic parameters to the same R&D-intensity and instantaneous 2017 output-growth targets. Lower discount rates put more weight on future growth and therefore raise the welfare value of R&D interventions. The main scenario ranking is preserved across $\rho \in \{5\%, 7.5\%, 10\%\}$; the table also reports the associated finite-value margins.

Appendix C Data, Implementation, and Numerical Checks

C.1 R&D Effort, Social-to-Private Marginal Returns, and Network Centrality

Table 13 reports descriptive cross-sectional checks that relate log model-implied and observed R&D effort and the log social-to-private marginal R&D return ratio to simple

Table 12: Discount-Rate Sensitivity with Recalibrated Dynamic Parameters

ρ	μ	δ	Min. stability $\rho - 2g_{\max}$	CM	CS	MM	SS	Opt. sub. (%)	Opt. sub. welfare
					Welfare (CC=100)				
5.0	0.0270	0.0099	0.0033	98.26	101.81	95.66	108.41	46	101.49
7.5	0.0398	0.0105	0.0335	98.71	100.71	95.60	108.45	36	100.64
10.0	0.0524	0.0112	0.0559	98.91	100.50	95.66	108.97	33	100.45

Notes: For each discount rate, μ and δ are recalibrated in the competitive 2017 equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 g_Y of 1.5%. Welfare columns are normalized to the competitive equilibrium at the same ρ. Min. stability is the minimum of $\rho - 2g_{\max}$ across solved main scenarios, where $g_{\max}$ is the largest real eigenvalue of the closed-loop transition matrix Φ; positive values indicate finite discounted values. Opt. sub. is the welfare-maximizing uniform R&D subsidy on the 0–50% grid, and opt. sub. welfare is normalized to no-subsidy competition at the same ρ.

Table 13: R&D Effort, Social-Private Returns, and Network Centrality

	Model log R&D	Observed log R&D	log(SMR/PMR)
Log specifications			
$\log z$	3.34***	3.64***	0.541***
	(0.087)	(0.137)	(0.076)
Product-market centrality	−1.75***	−0.073	−0.532***
	(0.061)	(0.089)	(0.078)
Technology-spillover centrality	0.250***	0.450***	0.837***
	(0.068)	(0.125)	(0.077)
Observations	739	715	739
R^2	0.873	0.715	0.251

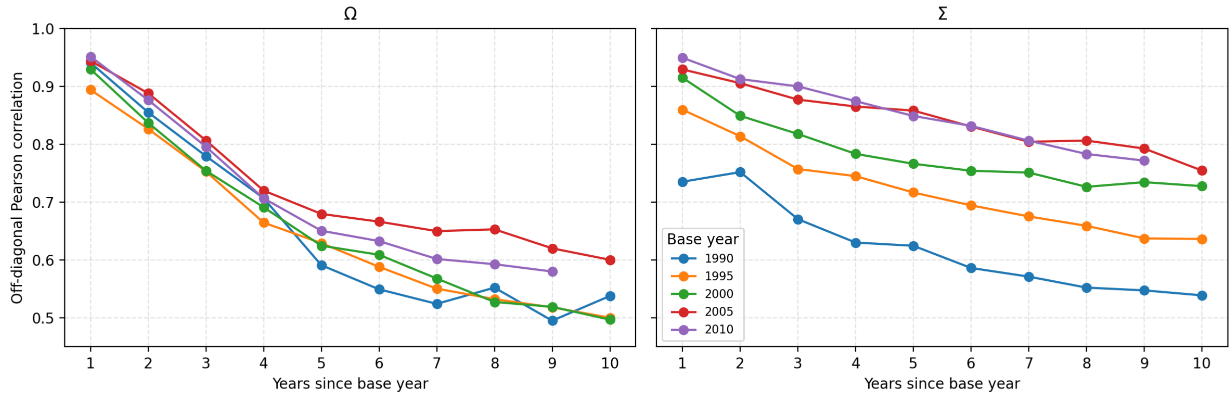
Notes: Cross-sectional regressions at the observed 2017 state for firms with positive PMR and SMR. Outcomes are log model R&D, log $\sqrt{\text{Compustat R\&D}}$, and log(SMR/PMR), the social-to-private marginal R&D return ratio. Centralities are max-normalized left-eigenvector measures; observed R&D must be positive. HC1 robust standard errors are in parentheses; intercepts are omitted. *, **, and *** denote significance at 10%, 5%, and 1%.

product-market and technology centrality measures at the observed 2017 state. The R&D columns provide reduced-form validation for the broad network forces in the model, while the return-ratio column summarizes model-implied wedge heterogeneity. Conditional on firm knowledge, the social-to-private return ratio rises with technology-spillover centrality and falls with product-market centrality.

The positive association between relative firm knowledge and the social-to-private return ratio is consistent with the model's static markup mapping. In the Cournot allocation, markup is increasing in the firm's equilibrium quantity share, and quantities satisfy $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$. In the estimated 2017 economy, the correlation between $\log z_i$ and the Lerner index is 0.889, summarizing the model-implied cross-sectional association.

The return-ratio coefficient also admits an exact numerator–denominator interpretation.

Figure 9: Persistence of Product-Market and Technology Networks



Notes: Each line fixes a base year and plots the Pearson correlation between the off-diagonal entries of the base-year matrix and the matrix h years later. Each comparison restricts the sample to firms observed in both years. The left panel reports the technology-spillover exposure matrix; the right panel reports the product-rivalry matrix.

Because $\text{PMR}_i = 2x_i$ under the private R&D first-order condition, the first-column coefficient on $\log z_i$, 3.342, is also the coefficient in the $\log \text{PMR}_i$ regression. Using the same 739 firms and regressors, the corresponding coefficient in the $\log \text{SMR}_i$ regression is 3.883, so their difference produces the return-ratio coefficient, $3.883 - 3.342 = 0.541$. Thus, both marginal returns rise strongly with knowledge, but the social return rises slightly faster. This reconciles the relative-wedge regression with the level decomposition above: although APP rises in levels, PMR also grows rapidly, while the approximately flat negative RP component becomes small relative to PMR. The positive gradient in the log return ratio therefore reflects the joint movement of APP, PMR, and RP.

C.2 Network Persistence

Figure 9 reports a panel diagnostic for the persistence of the two network primitives. For each base year, we compare the off-diagonal entries of the base-year matrix with the corresponding entries h years later, using firms observed in both years. The technology-spillover and product-rivalry matrices are highly persistent at short horizons, but the correlations decline with horizon length. This pattern supports observed-state comparisons around the 2017 networks.

C.3 Computational Scale

For the competitive dynamic problem, we solve the coupled system of Riccati equations by iterating on the coefficient matrices of firms' quadratic value functions. The implementation

Table 14: State Representation and Computational Scale

Model	State representation	Per-update arithmetic	Number of firms
Cavenaile et al. (2026)	Joint grid with k^n points	—	Up to 4
Our model	Continuous state; $O(n^3)$ stored coefficients	$O(n^4)$ per dense update	418–802

Notes: The state-representation column reports the size or storage requirement of the value representation. The per-update column separately reports arithmetic for our dense Riccati iteration; the corresponding quantity is not reported for Cavenaile et al. (2026).

stores the firm-level value matrices as a three-dimensional array, recomputes the policy matrix $\widetilde{\boldsymbol{X}}$ and closed-loop transition matrix $\boldsymbol{\Phi}$ at each iteration, and updates firm pages using blockwise vectorized matrix products. For the monopoly and social-planner dynamic problems, the corresponding dynamic block is a single Riccati equation because one decision maker internalizes all R&D choices. After solving the R&D rule, we compute household welfare and aggregate producer value from continuous-time Lyapunov equations under the induced closed-loop law of motion.

When the number of firms is n, each firm has an $n \times n$ matrix of value-function coefficients, $\boldsymbol{X}^i$. The competitive dynamic problem therefore stores one matrix page per firm, or n^3 coefficients in total. In the 2017 quantitative baseline, $n = 757$, so this amounts to about 434 million coefficients. The main arithmetic cost comes from Riccati updates such as $\boldsymbol{\Phi}^T \boldsymbol{X}^i$ and $\boldsymbol{X}^i \boldsymbol{\Phi}$: each dense matrix product costs order $O(n^3)$ for one firm page, and updating all n pages gives an $O(n^4)$ operation.

Within existing endogenous growth frameworks, solving a dynamic oligopoly problem involving all publicly traded U.S. firms with patents is computationally challenging. A joint grid with k points for each of n firm states contains k^n points. For instance, Cavenaile et al. (2026) analyze oligopolies with up to four firms using a discrete-state approach. Our continuous-state formulation avoids a joint state grid; it stores $O(n^3)$ value-function coefficients and requires $O(n^4)$ arithmetic per dense Riccati update. Table 14 separates these features of state representation from per-update arithmetic.

C.4 Riccati Solver, Selection, and Numerical Checks

For the competitive dynamic problem, the numerical algorithm used in the reported quantitative exercises iterates on the coefficient matrices of firms' quadratic value functions in the deterministic specification, $\gamma = 0$. Let k denote the iteration index. Given the current firm-level value matrices $\boldsymbol{X}_k^1, \dots, \boldsymbol{X}_k^n$, we recover $\widetilde{\boldsymbol{X}}_k$, construct the implied closed-loop matrix $\boldsymbol{\Phi}_k$, and update the matrices using the forward-Euler step

$$\frac{\boldsymbol{X}^i_{k+1} - \boldsymbol{X}^i_k}{\Delta} = \boldsymbol{Q}_i - \mu^2 \boldsymbol{X}^i_{i,k} \left(\boldsymbol{X}^i_{i,k}\right)^T + \left(\boldsymbol{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^i_k + \boldsymbol{X}^i_k \left(\boldsymbol{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right).$$

Here Δ denotes the pseudo-time step for the fixed-point iteration. The vector $\boldsymbol{X}^i_{i,k}$ is the ith column of firm i's value-function matrix at iteration k, equivalently the transpose of the ith row of $\widetilde{\boldsymbol{X}}_k$. The stochastic Riccati equations in the model include the diffusion term shown in Equation (11); the reported solver diagnostics evaluate the deterministic system used in the quantitative analysis. The implementation follows these steps:

1. Initialize $\boldsymbol{X}^i_0 = \boldsymbol{0}$ for all firms.
2. After each update, recover $\widetilde{\boldsymbol{X}}_k$ from the firm-level value matrices and construct the closed-loop transition matrix $\boldsymbol{\Phi}_k$.
3. Update the competitive firm-value matrices using the equation above.
4. Stop when the maximum absolute change in $\widetilde{\boldsymbol{X}}_k$ falls below the convergence tolerance, and evaluate stability using the largest real eigenvalue of $\boldsymbol{\Phi}_k - \rho\boldsymbol{I}/2$.
5. For the monopoly and social-planner benchmarks, solve the corresponding single-agent Riccati equation with the same convergence criterion, using the path-flow drift term $\boldsymbol{X}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T\boldsymbol{X}$.

The initialization also fixes the equilibrium-selection rule for the coupled Riccati system. For the no-subsidy competitive solves, the algorithm starts from zero value-function coefficients, $X^i_0 = \boldsymbol{0}$, and reports the stabilizing fixed point reached by the pseudo-time Riccati flow. This selected solution can be interpreted as the limit of finite-horizon LQ feedback games with zero terminal values when that limit exists (Cartigny and Michel, 2003). In the uniform-subsidy exercise, continuation over the subsidy grid preserves this branch by using each grid-point solution to initialize the next.

Table 15 reports scale-adjusted Riccati residuals and deterministic stability margins for the selected solutions.

C.5 Non-Negativity Checks

This subsection evaluates the closed-form static rules, $\boldsymbol{q}^u(\boldsymbol{z}) = \boldsymbol{N}\boldsymbol{z}$, and linear R&D policies, $\boldsymbol{x}^u(\boldsymbol{z}) = \mu\widetilde{\boldsymbol{X}}\boldsymbol{z}$, at the observed 2017 knowledge-capital vector. The reported projections

Table 15: Riccati Equation and Stability Diagnostics

Scenario	Riccati residual (relative max norm)	$\max \mathrm{Re}(\Phi - \rho I/2)$	$\rho - 2g_{\max}$
CC	1.7×10^{-6}	-0.0375	0.0751
CM	1.8×10^{-6}	-0.0369	0.0738
CS	1.6×10^{-6}	-0.0319	0.0638
MM	1.4×10^{-6}	-0.0368	0.0735
SS	1.8×10^{-6}	-0.0279	0.0559
CC_s_33	1.6×10^{-6}	-0.0342	0.0684

Notes: The Riccati residual is the scale-adjusted max norm of the deterministic algebraic equation evaluated at the reported solution. $g_{\max}$ is the largest real eigenvalue of Φ. Negative values of $\max \mathrm{Re}(\Phi - \rho I/2)$, equivalently positive $\rho - 2g_{\max}$, indicate the deterministic stabilizing condition used in Assumption 1. All reported rows satisfy this condition.

hold the solved dynamic policies fixed while imposing non-negativity on the evaluated controls.

The main scenario table evaluates R&D resource cost, instantaneous 2017 output growth, and welfare under the same solved policy.

Static Quantity Checks

Table 16 reports negative static quantities at the observed 2017 knowledge-capital vector together with the corresponding static nonnegative-quantity QPs. The competitive static mapping (CC, CM, and CS) yields positive quantities. In the full static reallocation benchmarks, 69 firms in MM and 234 firms in SS have negative quantities, accounting for 0.07 and 0.79 percent of the quadratic quantity aggregate. At the same state and dynamic policy, the static QPs move output from 96.33 to 96.27 in MM and from 109.29 to 108.82 in SS.

R&D Effort Checks

Table 17 reports the number and severity of negative R&D effort at the observed 2017 knowledge-capital vector, together with the growth effect of projecting negative effort to zero. The competitive benchmark has positive R&D effort for every firm. Negative effort appears in some counterfactuals, especially CM, where 421 firms have negative effort with a negative R&D cost share of 5.11 percent; the other counterfactuals have much smaller negative R&D cost shares. Under the fixed solved policy, setting $\boldsymbol{x}^{+}(\boldsymbol{z}) = \max\{\boldsymbol{0}, \boldsymbol{x}^{u}(\boldsymbol{z})\}$ componentwise in both R&D expenditure and the knowledge drift changes instantaneous 2017 g_Y by less than 0.003 percentage points in every scenario.

Table 16: Negative Quantities and Nonnegativity-Constrained Output at the Observed 2017 State

Scenario	Neg. firms	Neg. q^2 (%)	Closed-form output	$q \geq 0$ QP output	Output change	Zero-quantity firms
MM	69/757	0.07	96.33	96.27	-0.06	77/757
SS	234/757	0.79	109.29	108.82	-0.47	283/757

Notes: The table evaluates the closed-form static mapping $\boldsymbol{q}^u(\boldsymbol{z}) = \boldsymbol{N}\boldsymbol{z}$ at the observed 2017 knowledge-capital vector. Negative quantities arise only in the full static reallocation benchmarks MM and SS; the competitive static mapping (CC, CM, and CS) yields positive quantities. Negative q^2 is $\sum_{i:q_i^u<0}(q_i^u)^2/\sum_i(q_i^u)^2$. Firms with negative quantities account for at most 1.03 percent of observed sales and 2.58 percent of observed R&D. The QP columns solve the corresponding static nonnegative-quantity problem at the same state with the dynamic policy and Riccati controller held fixed; output is normalized to CC=100.

Table 17: Negative R&D Effort and Projected-Positive Growth at the Observed 2017 State

Scenario	Neg. firms	Neg. R&D cost (%)	Sales (%)	Obs. R&D (%)	Projected Δg_Y (pp)
CC	0/757	0.00	0.00	0.00	0.0000
CM	421/757	5.11	5.27	6.93	0.0014
CS	23/757	0.01	0.01	0.07	-0.0001
MM	228/757	0.95	1.36	1.68	-0.0007
SS	111/757	0.17	0.27	0.72	-0.0029

Notes: Negative R&D cost share is $\sum_{i:x_i<0} x_i^2/\sum_i x_i^2$ under the solved linear policy at the observed 2017 state. Sales and observed R&D are shares of observed 2017 totals accounted for by firms with negative R&D effort. Projected Δg_Y is the change in instantaneous 2017 g_Y when the projected-positive policy $\boldsymbol{x}^+(\boldsymbol{z}) = \max\{\boldsymbol{0}, \boldsymbol{x}^u(\boldsymbol{z})\}$ is used consistently in both R&D expenditure and the knowledge drift, with the solved policy held fixed.

Transition-Path Checks

Table 18 extends the sign check along deterministic closed-loop paths from the observed 2017 state and reports the welfare effect of applying the projected-positive policy along each path. Knowledge capital remains positive in all scenarios over the 50-year horizon. In the competitive benchmark, the first negative R&D effort appears after 24.5 years and the first negative static quantity after 33.3 years. The constrained and full-control counterfactuals have some negative controls or quantities at the observed state, but the corresponding negative mass is small relative to the aggregate objects in the scenario table. Projected-positive welfare differs by at most 0.13 percentage points, and the scenario ranking is unchanged.

Table 18: Transition-Path Sign Checks and Projected-Positive Welfare Robustness

Scenario	First $x < 0$ (years)	First $q < 0$ (years)	Max neg. R&D cost (%)	Max neg. q^2 (%)	Projected Δwelfare (pp)
CC	24.5	33.3	0.004	0.002	0.0001
CM	0.0	30.9	6.617	0.004	0.1321
CS	0.0	23.9	0.084	0.011	-0.0005
MM	0.0	0.0	0.954	0.125	0.0196
SS	0.0	0.0	0.764	1.086	-0.0115
33% subsidy	15.4	26.9	0.012	0.007	0.0001

Notes: The sign columns simulate deterministic closed-loop transition paths for 50 years from the observed 2017 knowledge-capital vector; the 33% subsidy row uses the solved welfare-maximizing uniform-subsidy equilibrium. Knowledge capital remains positive in every scenario. A value of 0.0 in a first-violation column means the sign violation is present at the observed state. Negative R&D cost is $\sum_{i:x_i<0} x_i^2/\sum_i x_i^2$, and negative q^2 is defined analogously. Projected Δwelfare is the change in welfare, in percentage points of the CC value, when negative R&D effort under the solved feedback policy is set to zero and the resulting policy is used consistently along the path; the calculation integrates household payoffs for 100 years and adds a fixed-active-set terminal tail whose share is below 0.043 percent in every row. The projected-positive calculation holds the solved feedback policy fixed along the path.

C.6 BGP Positivity Checks

This subsection evaluates the long-run BGP properties of the solved closed-loop systems. The main counterfactuals evaluate welfare and g_Y at the observed 2017 knowledge-capital distribution.

Baseline BGP Positivity Check

For the long-run BGP, the dominant eigenpair of the unconstrained closed-loop system solves

$$\boldsymbol{\Phi}\bar{z} = g\bar{z}, \qquad \boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}. \tag{48}$$

Although $\boldsymbol{\Omega}$ is nonnegative under the row-receiver spillover convention, $\widetilde{\boldsymbol{X}}$ contains strategic R&D policy feedbacks and can have negative off-diagonal elements. Hence $\boldsymbol{\Phi}$ is not generally a Metzler matrix and need not preserve the positive orthant; a positive dominant eigenvector is therefore not guaranteed by Perron–Frobenius arguments. After orientation toward the observed knowledge-capital vector and normalization by $\sum_i |\bar{z}_i| = 1$, $\bar{z}$ is the candidate BGP distribution. A positive BGP requires $\bar{z}_i > 0$ for every firm. Table 19 reports the number and mass of violations in the five counterfactual scenarios.

Table 19: Positive-State Feasibility of the Unconstrained Dominant BGP

Scenario	Neg. $\bar{z}$	Neg. $\lvert\bar{z}\rvert$ share (%)
CC	508/757	47.6
CM	410/757	49.4
CS	356/757	49.5
MM	443/757	49.3
SS	297/757	50.9

Notes: The dominant eigenvector is oriented toward the observed 2017 knowledge-capital vector and normalized by $\sum_i |\bar{z}_i| = 1$. Neg. $\bar{z}$ counts negative entries; Neg. $|\bar{z}|$ share reports their absolute mass.

Technology Spillover-Floor Check

The baseline BGP feasibility results show that the unconstrained dominant eigenvectors are mixed-sign in all five 2017 scenarios. To gauge whether this long-run sign issue is driven by the sparse allocation of empirical spillover links, we compute one perturbation for the competitive benchmark. The perturbation preserves each recipient firm's total spillover exposure but spreads a fraction λ uniformly across other source firms:

$$\Omega_{ij}^{(\lambda)} = (1-\lambda)\Omega_{ij} + \lambda \frac{\sum_{k\neq i} \Omega_{ik}}{n-1}, \qquad j \neq i, \tag{49}$$

with the diagonal unchanged. For $\lambda = 0.15$, the oriented dominant eigenvector of the competitive benchmark is positive, while the perturbation changes the instantaneous drift by only about 2.7 percent in relative ℓ_2 norm and leaves deterministic transitions from the observed 2017 state nearly unchanged: over a 50-year horizon, the cross-firm correlation of knowledge-growth rates with the baseline path stays above 0.996 and aggregate R&D remains within 1 percent of the baseline path, while the continuation value at the observed state, $V(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X} \boldsymbol{z}_0$, differs by only 0.4 percent. This exercise is not a re-estimated model and is not used in the main counterfactuals; it only checks that the observed-state comparison is not mechanically driven by the mixed-sign asymptotic BGP.